

MY457: Reappraisal of Tellez, 2022

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1 Introduction

In this study, I will critically reappraise Tellez’s (2022) paper on the relationship between municipal-level displacement and palm-oil plantations in Colombia between 1993 and 2005. This study will start by discussing the paper’s empirical strategy, their data and their subsequent operationalisation. Then, I will discuss the identification assumptions such empirical strategy requires for causal identification, and some limitations that the author’s current approach presents, some of which were not discussed in the original paper. Following such discussion, this study will present some modifications to the author’s original empirical strategy. Following this, a replication of Tellez’s (2022) main results will be presented, accompanied by a set of robustness tests and a discussion of the findings, relative to the paper’s original interpretations.

2 Research Design

2.1 Tellez’s (2022) Empirical Strategy & Data

The paper’s main hypothesis is that palm-oil plantations led to higher municipal-level displacement rates given it is a land-intensive yet not-labour-intensive crop, so it requires of land but not people to cultivate or harvest it. He presents his theory in contrast to illegal crops (e.g., coca), which do not require of large masses of land, but require more people to work on them. The paper’s secondary hypothesis is that this plantation-induced displacement is highest when AUC groups are present, as they are more likely to collude with local (rural) elites than left-wing guerrilla groups (FARC). Tellez (2022) uses the first year in which a municipality started planting palm-oil as the year that a municipality first gets treated, and explores the effect this had on the rate of forced displacement (i.e., “*forced to leave their home as a result of an armed conflict, generalised violence, or human rights violations*”).

His treatment is a binary variable which equals 1 the first year a municipality gets its first palm-oil tree (and thereafter), or 0 otherwise. Tellez (2022) interacts his binary treatment with the total palm-oil production (logged) in the country per year, to account for the varying intensity of production at different years. The outcome is the rate of displaced people per 1,000 by municipality and year, though the paper applies an inverse-sine transformation to it (the author claims that the main model is robust to instead taking the natural log or using raw displacement numbers and controlling for population in a count model). Tellez (2022) includes two controls in his specification: a binary variable of whether a municipality has coca plantations; and the number of left-wing guerrilla attacks. In his models he includes country and year fixed-effects, estimated with the least squares dummy variables (LSDV) estimator. The simple model is represented in equation (1), while the interacted model is represented in equation (2). The interacted models do not have an individual coefficient for the interacted variables given they get dropped from the regression due to multicollinearity with the fixed-effects.

$$dispRate_{it} = palmOil_{it}\beta + \mathbf{X}'_{it}\boldsymbol{\gamma} + \alpha_i + \omega_t + \epsilon_{it} \quad (1)$$

Where:

$$palmOil_{it}\beta = \sum_{r \neq 0} \mathbf{1}[R_{it} = r] \tau_r$$

Meaning that the *palmOil* variable and its coefficients are relative to the treatment period of a unit. When the author presents its results, these relative-period treatment effects are aggregated together to create a single coefficient. This is the case for all equations below as well.

$$dispRate_{it} = palmOil_{it}\beta + natlProd * palmOil_{it}\phi + \mathbf{X}'_{it}\gamma + \alpha_i + \omega_t + \epsilon_{it} \quad (2)$$

The second-most important hypothesis of the paper is that this effect of palm-oil plantations on displacement rates is conditional on group presence. The paper presents two models which interact with the presence of FARC and paramilitary groups (AUC). The reasoning for this is that FARC were not really in contact with local economies, and were ideologically opposed to local (rural) elites. On the contrary, paramilitary groups were economically tied to localities and rural elites. While both were involved in drug trafficking and massacres, the author hypothesises that FARC were not being “opportunistic” through the forced displacement they cause, so plantations in the municipalities they control are uncorrelated with displacement rates, while paramilitary groups were opportunistic, and had an incentive to displace people to take up their land and plant palm-oil, in collusion with local (rural) elites. Their model is represented in equation (3).

$$dispRate_{it} = palmOil_{it}\beta + FARC/AUC * palmOil_{it}\delta + \mathbf{X}'_{it}\gamma + \alpha_i + \omega_t + \epsilon_{it} \quad (3)$$

2.2 Identification Assumptions

The paper’s causal identification strategy is a staggered difference-in-differences design. This identification strategy differs from the canonical difference-in-differences in the sense that now we allow for heterogeneous treatment effects conditional on relative time since treatment. This means that we assume that the effect of implementing palm-oil plantations differs, for example, five years since treatment with just one year since treatment. The main identification assumption from such a strategy is parallel trends, which assumes the following:

$$\mathbb{E}[dispRate_{i,t+1}(0) - dispRate_{i,t}(0)|G_i = 1, X_i = x] = \mathbb{E}[dispRate_{i,t+1}(0) - dispRate_{i,t}(0)|G_i = 0, X_i = x]$$

This can be read as assuming that the difference in potential outcomes in the post-treatment period (t+1) with the pre-treatment period (t) for units assigned to treatment (palm-oil plantations) and units assigned to control (no palm-oil trees) would have been the same, had none received treatment. In this case, as seen above, this is conditional on two covariates (number of FARC attacks, and whether there are coca plantations in the municipality). “Within” the parallel trends assumption lies another: no anticipation. This assumes that treated units do not react to treatment prior to it occurring. Given our data is yearly, for this assumption to be met, the displacement rate in the year prior to adoption has to be independent of taking up treatment the next year. The paper refers to these two assumptions as exogeneity of treatment (“*interventions are effectively random, conditional on time and group fixed effects*”), which, if met, often supports parallel trends and no anticipation.

Neither of these two identification assumptions can be directly tested for, as we never observe the potential outcome $\mathbb{E}[dispRate_{i,t+1}(0)|G_i = 1, X_i = x]$, but through an event-study plot we can test for *parallel pre-trends*, and visualise how differences in the outcome progress relative to the treatment period. Basically, we are assuming that the municipalities that had palm-oil planted would have followed similar trends in displacement rates as the municipalities that had no palm-oil planted, in the absence of treatment, as long as the two groups followed similar pre-trends in rates of displacement. The event-study plot, however, cannot be interpreted as proving that the parallel trends assumption is met or not, but rather serves as a visual falsification test. The event-study plot will assume a constant treatment effect across treatment

cohorts relative to treatment take-up, meaning those treated in 1995, 1999, and 2003 will be assumed to have the “same” treatment effect for 1996, 2000, and 2004, given it is just one year after treatment take-up, respectively. If we find no statistically significant deviations between treated and control, we can say that we find no evidence that the parallel trends assumption is not met.

Another implicit assumption in any causal inference strategy is the stable unit treatment value assumption (SUTVA). SUTVA here could be violated, given that it assumes there is no “spillover” from the treatment effect of one unit on the outcome of another. In the words of the paper, we are assuming that if a town starts planting palm-oil trees this would not increase or decrease the neighbouring town’s displacement rates. While we cannot really test for SUTVA, it is likely not met here, given that if a town has much of its population displaced given the palm-oil plantations, many will go to the neighbouring town, which will then likely experience some displacement itself given the intake of new people, despite not planting palm-oil trees themselves. This issue may also indicate potential anticipatory effects, as a town would likely know if the neighbouring town is planting palm-oil trees (or whether it is being subject to violent attacks from paramilitary groups), and may leave their town prior to the palm-oil tree plantations (or violent attacks) starting there too (i.e., a form of anticipatory bias). It may be that the bias this potential SUTVA violation creates is only downwards, but it is difficult to assess without properly checking whether this is true.

2.3 Limitations and Critique

Tellez’s paper presents multiple limitations, including those which signal a potential violation of the key identification assumptions in staggered difference-in-differences designs. Another major limitation of the paper is the treatment variable it uses. In the first model (eq.1), the author is assuming that the effect of palm-oil plantation on displacement rates is proportional across municipalities, regardless of the amount of palm-oil trees planted in each town or across the nation in general. In the second model (eq.2), it assumes that the *size* of palm-oil plantations is equal across all municipalities with palm-oil trees, as it accounts for differing rates of plantation *nationally*, but given the lack of a lower level of data it cannot really identify whether larger plantations drive larger displacement rates. In the third model (eq.3), it again assumes that the effect of palm-oil plantation on displacement rates is proportional across municipalities, regardless of the amount of palm-oil trees in each town or across the nation in general, as it stops interacting the treatment with the (logged) national production of palm-oil.

Another important critique I have of the paper is his weak explanation for why it includes coca plantations as a covariate in his regressions. In the introduction, he mentions how coca plantations rarely lead to a displacement of the population given it is a labour-intensive crop but not at all land-intensive (i.e., as it requires to be quite secretive), yet says he is controlling for it to account for “changes in the drug trade”, and it is not exactly clear why coca plantations would be a good control for this purpose. The lack of further explanations is quite confusing, as it is not clear whether he is controlling for it “just in case” it does have an effect on displacement rates (e.g., through drug trade), or for other reasons. Additionally, other theories exist which posit that coca does indeed drive displacement given Plan Colombia’s efforts, which forced displacement to eradicate coca plantations (Dion & Russler, 2008), which is a more reasonable justification.

In his results, he finds a statistically significant and substantively important effect of coca plantations on displacement rates (i.e., even larger than the effect of palm-oil plantations), which may favour the idea that he controls for it because, contrary to expectations, it does have an effect on displacement rates. Yet, he should have justified its use properly ex-ante viewing the results, such as through a DAG, which helps visualise the ways in which controlling for variables “just in case” can introduce bias in our estimates.

A big concern I have with the paper is his construction of the FARC and AUC dummies which interact with the treatment variable. There are four possible cases, as can be seen in the following cross-tabulation (Table 1). In his construction of the dummy variable, when multiplying the treatment with the AUC dummy, he is calculating the effect of a municipality having either AUC presence or *both* AUC and FARC presence versus just having FARC presence or no armed-group presence at all. The same applies to the FARC dummy.

Table 1: Cross-tabulation of AUC and FARC Presence by Municipality (Unique)

	FARC: No	FARC: Yes
AUC: No	112	292
AUC: Yes	54	660

An alternative, and possibly better, option would be to construct a categorical (*unordered*) variable which takes four values: 0 (baseline) for when neither FARC or AUC are present; 1 for when FARC is present but not AUC; 2 for when AUC is present but not FARC; and 3 for when both AUC and FARC are present. By interacting our treatment with this categorical variable, rather than two separate models with distinct dummy variables, we can better estimate the point-estimate differences between having no group presence, FARC, AUC, or both present (most likely scenario) on displacement rates, when interacted with palm-oil plantations.

The author notes at the end of the paper that he cannot confidently establish whether it is palm-oil plantation that drives displacement, or whether displacement precedes palm tree plantation. While it is understandable to be concerned about this, an easy “confirmation” of the reverse causality concern would be to run the same models, but exchanging the dependent (displacement) for the independent (palm-oil plantation) variable, and checking whether there exists at least a correlation between them. While this is just confirmatory, it does help visualise the problem, so further studies can ideate potential solutions.

2.4 Modifications

Instead of using the `feelm()` function I am using the `feols()` function from the ‘fixest’ package, and, given `stargazer()` does not support `feols()`, I have created the main tables with `texreg()` and `kable()`, a minor coding change from the original paper.

I have added a model (i.e., Model 4+) which includes the ‘`farc_attack`’ covariate to the regression and, as will be seen, the results are not sensitive to this addition. The original results plus Model 4+ will be reported as is. The author did not include the ‘`farc_attack`’ covariate in Model 4 for some reason he does not explain.

I have included two triple-interaction models (palm-oil plantation x FARC/AUC x national production of palm-oil), to account for the years during which palm-oil plantations grew in size particularly fast, to test if his secondary hypothesis holds under this change. See model in equation (4).

$$\begin{aligned} dispRate_{it} = & palmOil_{it}\beta + natlProd * palmOil_{it}\phi + FARC/AUC * palmOil_{it}\delta \\ & + natlProd * FARC/AUC * palmOil_{it}\kappa + \mathbf{X}'_{it}\gamma + \alpha_i + \omega_t + \epsilon_{it} \end{aligned} \quad (4)$$

As mentioned above in the limitations, I will include a model interacting with the new ‘`group_presence`’ variable. This should help us visualise further caveats within his secondary hypothesis that FARC presence, when interacted with palm-oil plantations, is uncorrelated with displacement rates. I will also present a model that interacts with both dummies *individually*, which will permit interpreting the four possible coefficients: when no group is present (base coefficient), and when either equals one but not the other. Due to multicollinearity, both the ‘`group_presence`’ interaction model and the one interacting with both dummies always drops one coefficient.

Alongside the paper’s TWFE estimation, I have estimated the ATT with various modern difference-in-difference estimators: Callaway & Sant’Anna’s (2021); Sun & Abraham’s (2021) with all three outcome types (original, logged, and inverse-sine); Liu, Wang & Xu’s (2024) FEct, IFect and MC (imputation methods) estimators; Gardner’s (2022) two-stage diff-in-diff; and Borusyak *et al.*’s (2024) own imputation estimator. This is useful given their slightly different approaches to deal with the TWFE limitations widely covered in the above literature, but quickly summarised as a problem with how TWFE makes “forbidden comparisons” between already-treated and not-yet-treated units when estimating the ATT, weighing many

different treatment effects, sometimes negatively, which is non-intuitive, can severely mislead, and is difficult to interpret.

3 Replication

The main results from the paper are those related to the municipal-level analysis of displacement rates. While the paper’s survey analysis is interesting qualitatively, and his land restitution analysis may be interesting to expand on, neither is done with a causal inference strategy, so replicating the results would not be an efficient use of the limited space here. I have replicated Figure 1 from his paper, which illustrates the difference in displacement rates between municipalities with palm oil plantations and those without palm oil plantations. The remaining figures in the paper involved a similar type of visualisations, but were not as intuitive nor related to the paper’s main hypothesis of palm-oil’s effect on municipal-level displacement rates, so I have kept them in the 6.

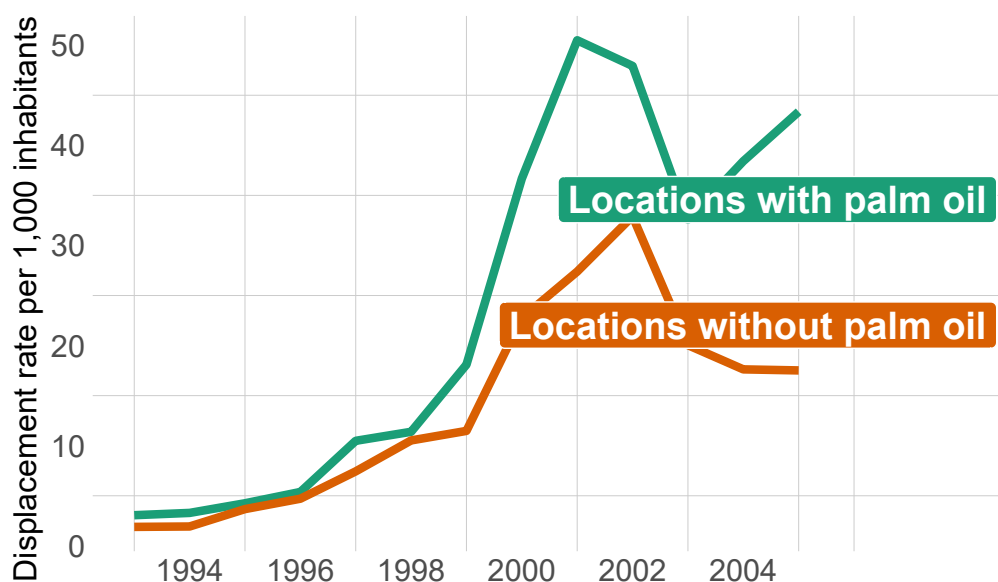


Figure 1: Rate of Displacement (per 1,000 residents) of Municipalities With and Without Palm-Oil Cultivations

Figure 1, shows the rationale for studying the relationship between palm oil plantations and displacement rates in Colombia, and doing so with a causal inference strategy such as difference-in-differences given the limitations a naïve difference in means estimation has (e.g., omitted variable bias). On the other hand, the paper does not present an overview or visualisation of how many municipalities have palm oil plantations, how the number of plantations grew over the sampled years, or whether they have missing data. In Figure 2 and 3, I show the trend in palm-oil plantations in Colombia and show how there are some municipalities which only have data from a specific year onwards. We can see how it is a minority of municipalities that have palm-oil plantations, so our treated group is much smaller than the control. This is not necessarily bad, but is important to know for the context.

Specified colors in the order of: Under Control, Under Treatment, Missing.

In Table 2, I provide a cross-tabulation of the presence of coca and palm oil plantations across the sample. Here we can see how unlikely it is that a municipality has both coca and palm oil plantations, which makes sense given the different environmental conditions each requires. This is important for the choice of controls, as it may indicate why the author controls for coca. Displacement rates in municipalities without palm oil

Plantation of Palm-Oil Across Municipalities

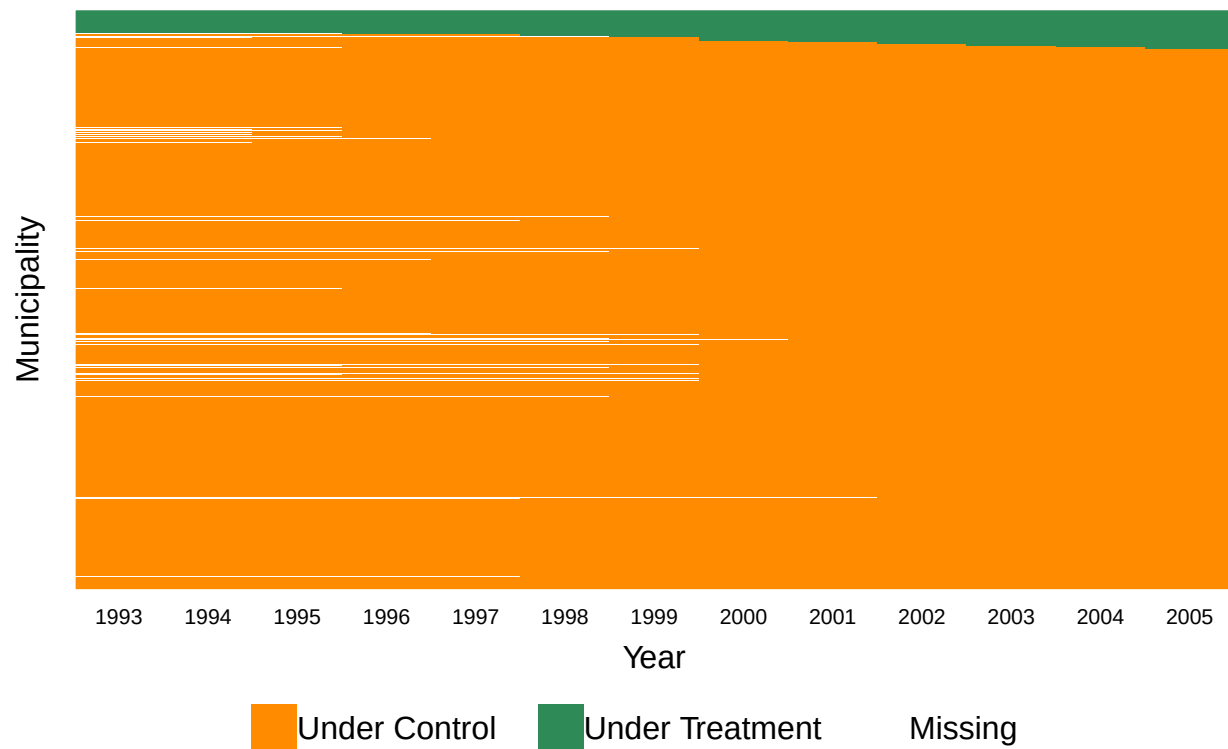


Figure 2: Plantation of Palm-Oil Across Municipalities.

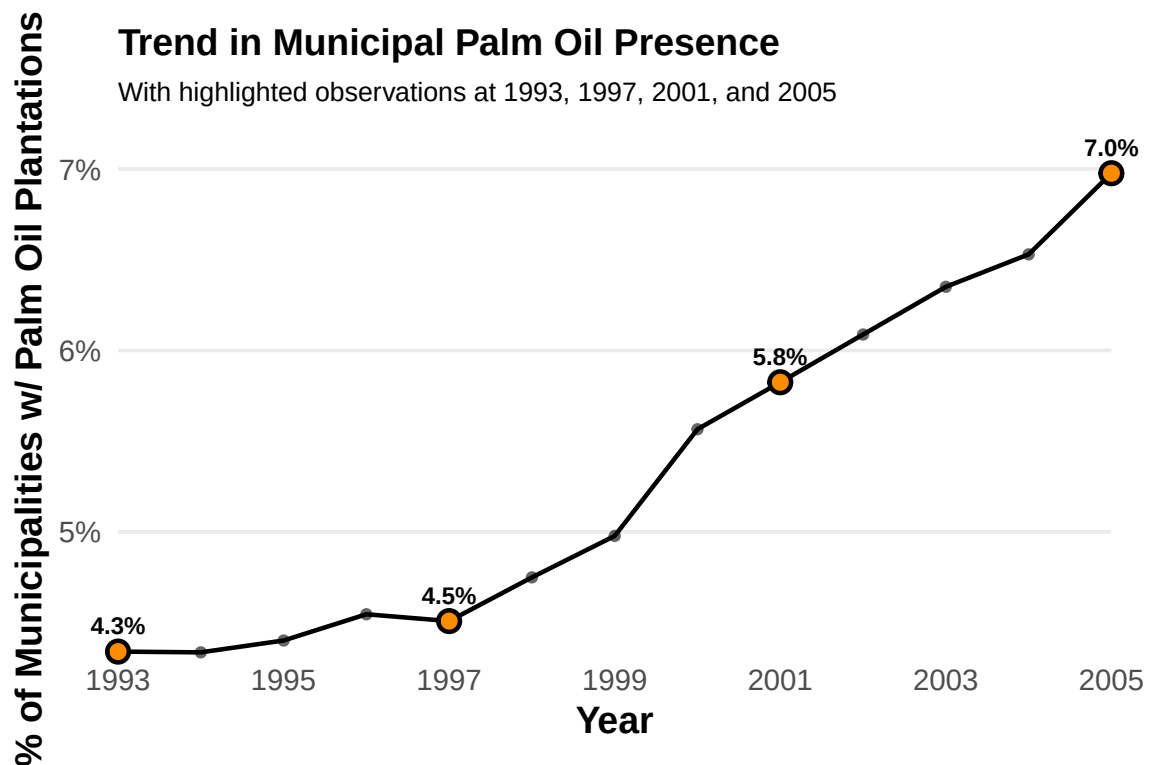


Figure 3: Trend in Municipal Palm Oil Presence.

Table 2: Cross-tabulation of Coca and Palm Plantations by Municipality (Unique)

	Palm Plantation: No	Palm Plantation: Yes
Coca Plantation: No	1069	71
Coca Plantation: Yes	260	30

(control) may be more directly affected by the drug trade than those with palm oil given the more likely presence of coca plantations, so by including it in the model he can attempt to control for the displacement effects of drug trade, obtaining a better estimate.

While the author only includes an event-study plot in the online appendix, here I present three event-study plots with the Sun & Abraham (2021) (interaction weighted) estimator, with the three possible outcomes one can test the effect of palm-oil plantations on municipal-level displacement rates. The inverse-sine transformed plot is exactly the same as the one reported in the online appendix, and seems to suggest parallel pre-trends are met. The same applies to the log-transformed outcome model. The non-transformed outcome model shows a statistically significant deviation in the outcome five years prior to treatment take-up, which was already present in the other models but was not statistically significant there. While this may be due to noise in the data (as there are no other statistically significant pre-trend violations), it may also be that indeed there is some consistent shock exactly five years prior to the first palm oil tree being planted that drives displacement rates up and violates parallel trends. This motivates the use of a placebo (pre-treatment period) test, which can be easily implemented with the ‘fect’ package.

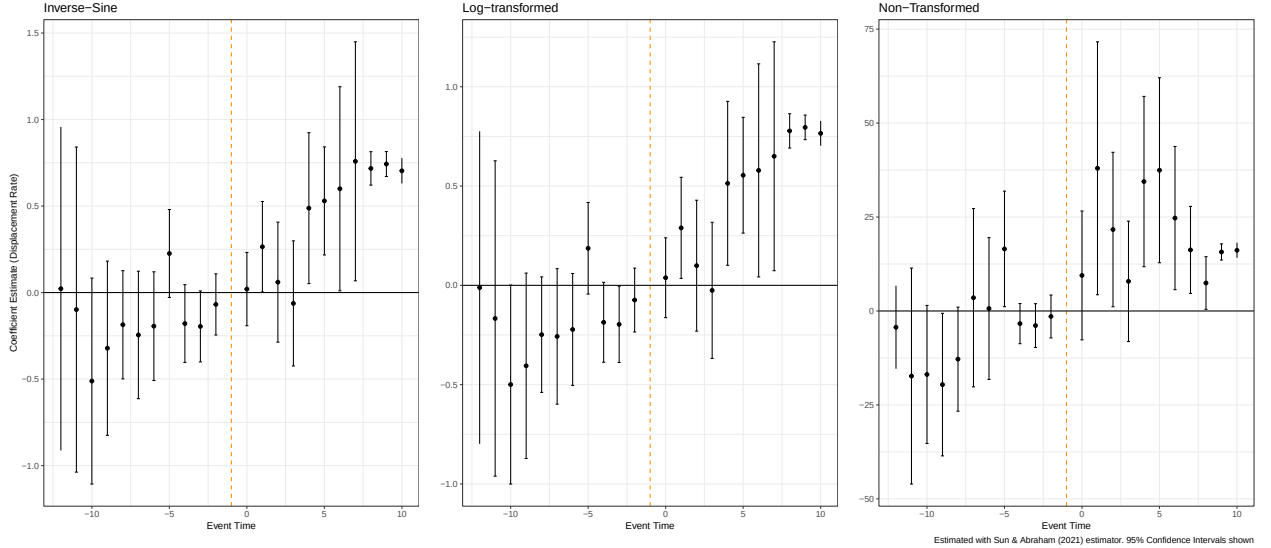


Figure 4: Event Study Plot for Displacement Rates (per 1,000) – Inverse-Sine, Log, and Non-Transformed Outcomes.

These placebo (pre-treatment period) tests estimate the treatment effect in a window before any intervention occurs ($t=0$), where the true effect should mechanically be zero, and serve as a falsification test of the parallel trends assumption. These tests are done with the FEct estimator, which while different from the interaction weighted (Sun & Abraham) estimator, is less robust to parallel trends than the IFect or MC methods proposed by Liu, Wang and Xu (2024), and is conceptually similar to a TWFE model, which is the estimator used in the paper’s main specification. These tests provide two values: a placebo test p-value, and a placebo equivalence test (TOST) p-value. The placebo test p-value, if lower than 5%, would reject the null hypothesis of no differences, signalling a violation of parallel trends. On the other hand, the TOST checks whether the pre-period is “flat” (if the pre-treatment coefficients are close enough to a chosen value which by default is 0.36 times the standard deviation of the outcome after demeaning), and if it is bigger than 5% we cannot reject the existence of some trivial pre-treatment deviations. The placebo test with reference at

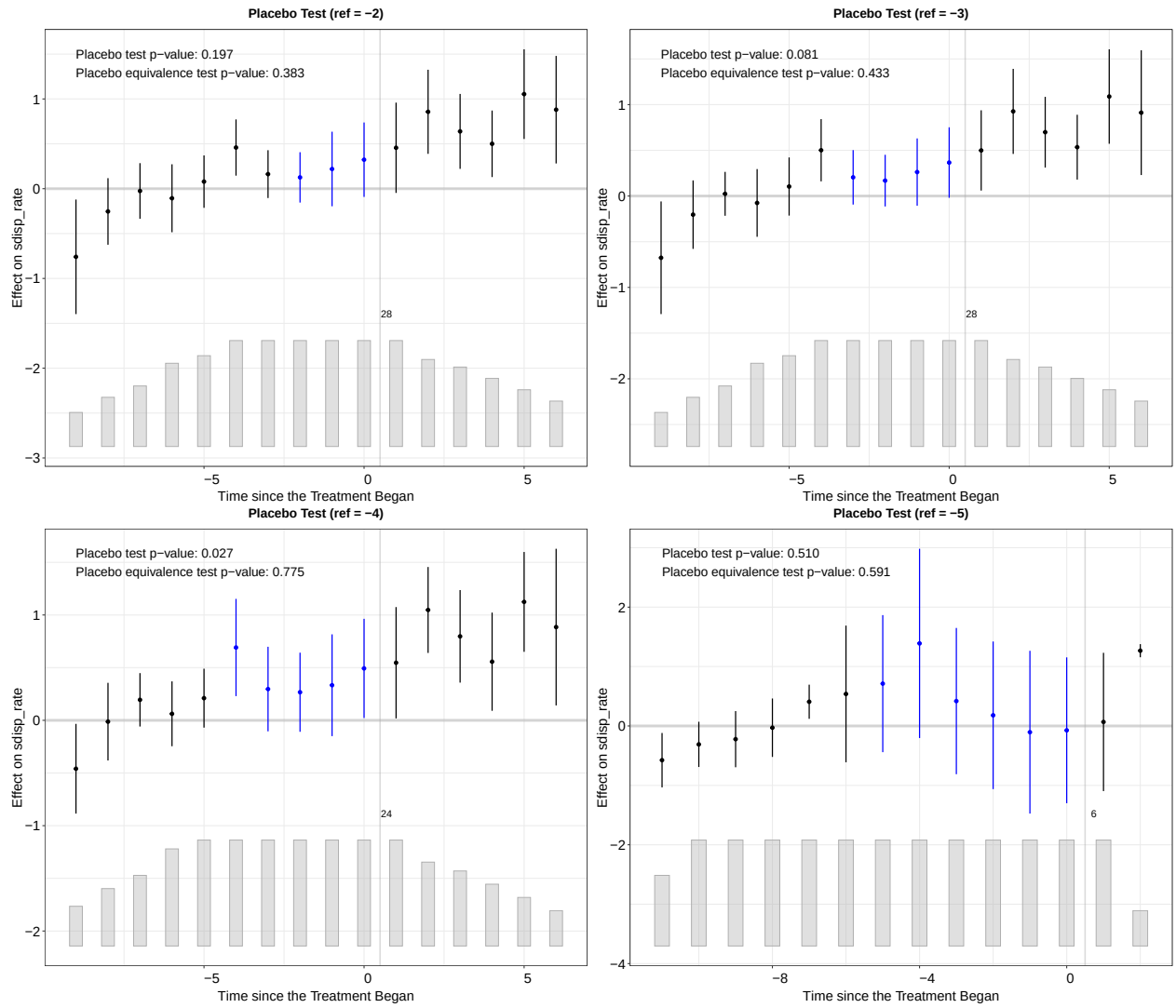


Figure 5: Placebo Tests for Parallel Trends (References = -2, -3, -4, -5).

$t=-4$ has a p-value lower than 5%, signalling that parallel trends may be violated, though the other placebo tests do not indicate the same. All placebo tests have a TOST p-value bigger than 5%. While these tests by themselves are not proof of the parallel trends assumption being met or not, their results call for caution in interpreting post-treatment effects, and motivate the use of Rambachan and Roth’s (2023) robust inference and sensitivity analysis tools, as it is possible (and even suggested) that the parallel trends assumption is violated.

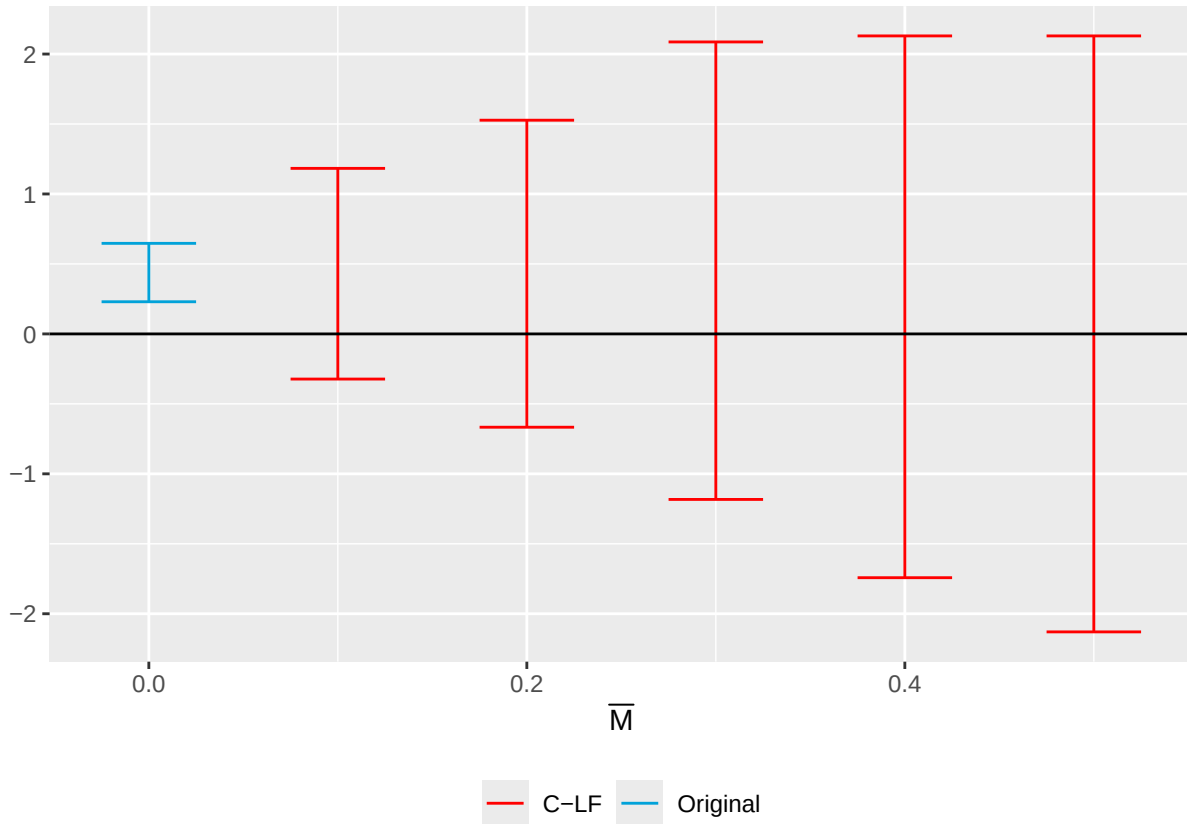


Figure 6: Sensitivity Analysis (Relative Magnitudes) – Rambachan and Roth (2023).

In Figure 6, we are seeing the original model’s (with inverse-sine transformation) aggregated ATT, and how it reacts to deviations (induced “bias”) from the parallel trends assumption, relative to the observed pre-treatment fluctuations (indexed by $\bar{M}$). As we can observe here, any minor fluctuations from our parallel trends assumption make the ATT statistically *insignificant*, so we cannot fully reject the null. Nonetheless, as Chiu *et al.* (2025) report, very few papers manage to remain statistically significant with slight deviations in the parallel trends assumption, so the fact that this paper’s results do not remain significant is not entirely surprising.

In Table 3, I present the models used by Tellez plus the one with the additional ‘farc_attack’ covariate, to illustrate that the results are fully replicable, and that including the additional covariate barely changes the estimates. In Table 4, I present my modified models, including the triple interactions models, the model interacted with my group presence variable, and the model interacted with the AUC and FARC dummies. All models presented use the inverse-sine transformed outcome, and their coefficients cannot be easily interpreted, yet their sign and magnitude still illustrate various interesting points not covered by Tellez. All models were calculated with the canonical TWFE estimator. While this is likely going to overestimate the ATT, further tests will be done to assess the validity of the coefficients.

The first curiosity I find from replicating the paper’s results is that the coefficient from coca is higher than the coefficient from palm-oil, directly contradicting his claim from the paper: “*the logic is borne out in*

	Model 1	Model 2	Model 3	Model 4	Model 4+
Palm-oil plantation	0.382** (0.121)	-2.056** (0.642)	-1.213*** (0.023)	0.692 (0.548)	0.713 (0.558)
FARC Attack	0.148*** (0.013)	0.148*** (0.013)	0.148*** (0.013)		0.148*** (0.013)
Coca plantation	0.541*** (0.056)	0.539*** (0.056)	0.541*** (0.056)	0.580*** (0.058)	0.542*** (0.056)
Plantation x Natl Prod		0.248*** (0.063)			
Plantation x AUC			1.614*** (0.120)		
Plantation x FARC				-0.327 (0.559)	-0.357 (0.570)
Num. obs.	14192	14192	14192	14208	14192
Num. groups: cod_dane	1118	1118	1118	1118	1118
Num. groups: year	13	13	13	13	13

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 3: Statistical Models (Replication of Téllez and Additional Specification)

anecdotal accounts that indicate displacement is relatively rare in illegal crops (e.g., coca)” (pp.404). While this is not exactly related to the causal inference problem faced here, it illustrates that the mechanism he proposed through which palm-oil should drive *more* displacement may not be entirely true.

In my modified models I find that, as expected, higher national production of palm-oil increases the displacement rate even when interacted with the AUC and FARC dummies. I also find that the municipal-level displacement rate is lowest when there is a plantation *and* FARC is present ($\hat{\beta} = -1.212 - 0.768 = -1.980$), which is not totally intuitive as one would have expected displacement rates to be lowest when there are no armed groups whatsoever. This effect may be due to data quality issues regarding armed group presence, as rightfully pointed out by the author, so should be interpreted carefully. Indeed, as the author rightfully pointed out, the highest displacement rates occur when *only* AUC is present ($\hat{\beta} = -1.212 + 2.337 = 1.125$).

Nonetheless, FARC presence *is* associated with higher displacement rates than when there are no armed groups when interacted with palm oil plantations, but only when paramilitary groups are also present, as can be seen from the ‘Plantation x Both’ coefficient ($\hat{\beta} = -1.212 + 1.569 = 0.357$). In Table 1, we saw how most often we find both armed groups present in the same municipality, so on most occasions the increased displacement rate from AUC presence likely also comes from the fact that both armed groups are in direct conflict for economic resources and territorial control. While it seems true that in AUC-only municipalities there is a higher displacement rate than in FARC-only ones when there are palm-oil plantations, it is not the case that FARC presence is totally unrelated with displacement rates as argued by the author, given the lack of statistical significance in Table 3’s Model 4 ‘Plantation x FARC’ coefficient.

Nonetheless, the estimates from these tables come from the TWFE estimator, and this estimator is limited in staggered difference-in-differences settings for the reasons explained in the Modifications section. While we have already presented the SA-estimated event-study plots, re-estimating the main model with other modern estimators is recommended (Chiu *et al.*, 2025). Here, I will make use of Liu, Wang and Xu’s (2024) counterfactual estimators, given their fundamentally different approach (especially IFect) to deal with TWFE’s “forbidden comparisons” and estimate the ATT.

The FEct estimator obtains the highest estimates of the three counterfactual estimators tested here, estimating an aggregated ATT that is even higher than the TWFE one presented in Table 3. The matrix completion (MC) estimator again finds an ATT larger than the TWFE estimator. Neither of these estimators finds a statistically significant deviation in the pre-treatment trends. Nonetheless, the IFect estimator presents a very different picture from these two models. First off, it identifies a statistically significant and substantially large deviation in the 5th year prior to treatment, signalling that parallel pre-trends is not met. Then, it

	Triple Interaction		Group Presence	Multiple Dummies
	Model 1	Model 2	Model 3 (ref=None)	Model 4
Palm-oil plantation	-3.106*** (0.386)	-2.679 (2.523)	-1.212*** (0.023)	-1.212*** (0.023)
FARC Attack	0.142*** (0.013)	0.141*** (0.013)	0.148*** (0.013)	0.148*** (0.013)
Coca plantation	0.497*** (0.055)	0.514*** (0.054)	0.542*** (0.056)	0.542*** (0.056)
Plantation x AUC	1.752* (0.766)		2.337*** (0.291)	2.337*** (0.291)
Plantation x Natl Prod	0.197*** (0.039)	0.384 (0.227)		
AUC x Natl Prod	0.238*** (0.032)			
Plantation x AUC x Natl Prod	-0.029 (0.076)			
Plantation x FARC		0.935 (2.609)		-0.768* (0.313)
FARC x Natl Prod		0.410*** (0.037)		
Plantation x FARC x Natl Prod		-0.177 (0.236)		
Plantation x Both			1.569*** (0.119)	
Num. obs.	14192	14192	14192	14192
Num. groups: cod_dane	1118	1118	1118	1118
Num. groups: year	13	13	13	13

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 4: Statistical Models (Modified Specifications)

estimates a negative, and statistically insignificant, aggregated ATT, likely because it tries to correct for the parallel trends violation through its additional latent factor variable.

Given the variability in the IFect estimator, and the range of estimations that we have found throughout, I present in Figure 8 a graph with all the aggregated coefficients, including three estimators which were not presented before: Callaway & Sant’Anna’s (2021); Gardner’s (2022); and Borusyak *et al.*’s (2024). While Callaway & Sant’Anna’s estimator is similar in certain aspects to the Sun & Abraham (2021) estimator, it differs on others, and indeed estimates a different ATT, which signals less statistical significance than previous estimations. The latter two estimators (two-stage diff-in-diff and DIDimpute) are two alternative counterfactual (imputation-based) estimators (methods), largely similar to FEct (Chiu *et al.*, 2025), but producing slightly different estimates. Still, their estimations are statistically significant and support the paper’s results. Overall, these results point towards a positive, yet somewhat sensible effect of palm-oil plantations on displacement rates, *only* if parallel trends hold (as could be seen in Figure 6 with Rambachan and Roth’s sensitivity analysis).

4 Discussion

Overall, this reappraisal has found a few interesting limitations which, while not overturning the paper’s results, call for caution in interpreting these ATT estimates as causal. While the positive ATT from the main results table seemingly holds under various specifications and estimators (see Figure 8), the results were contested in various ways (e.g., high TOST p-values, assumption-sensible results, IFect’s negative ATT).

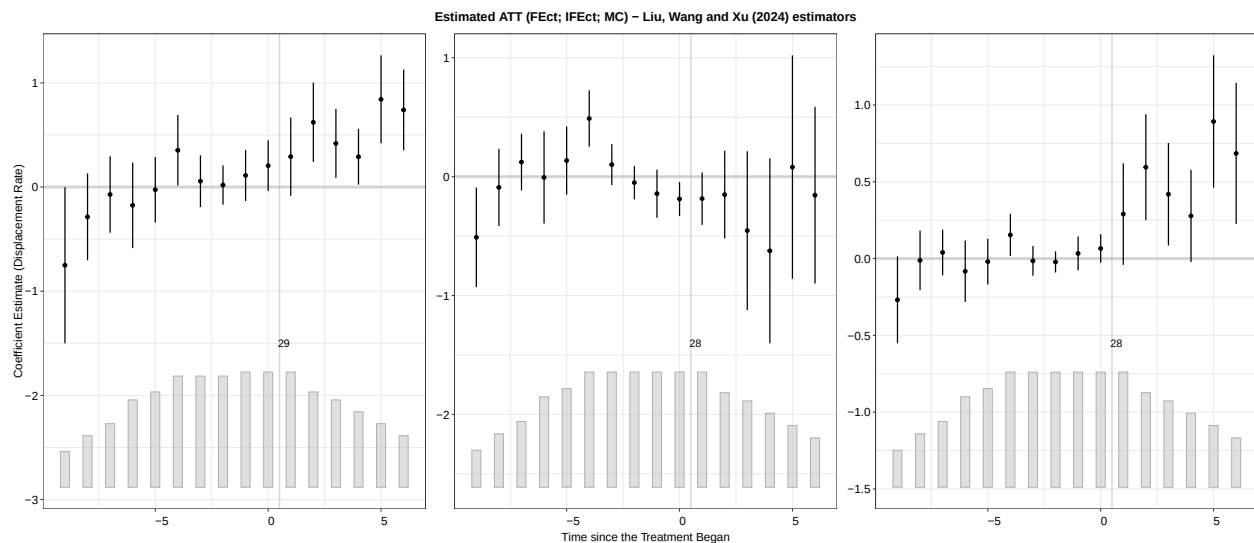


Figure 7: Estimated ATT (FEct, IFect, and MC) – Liu, Wang and Xu (2024).

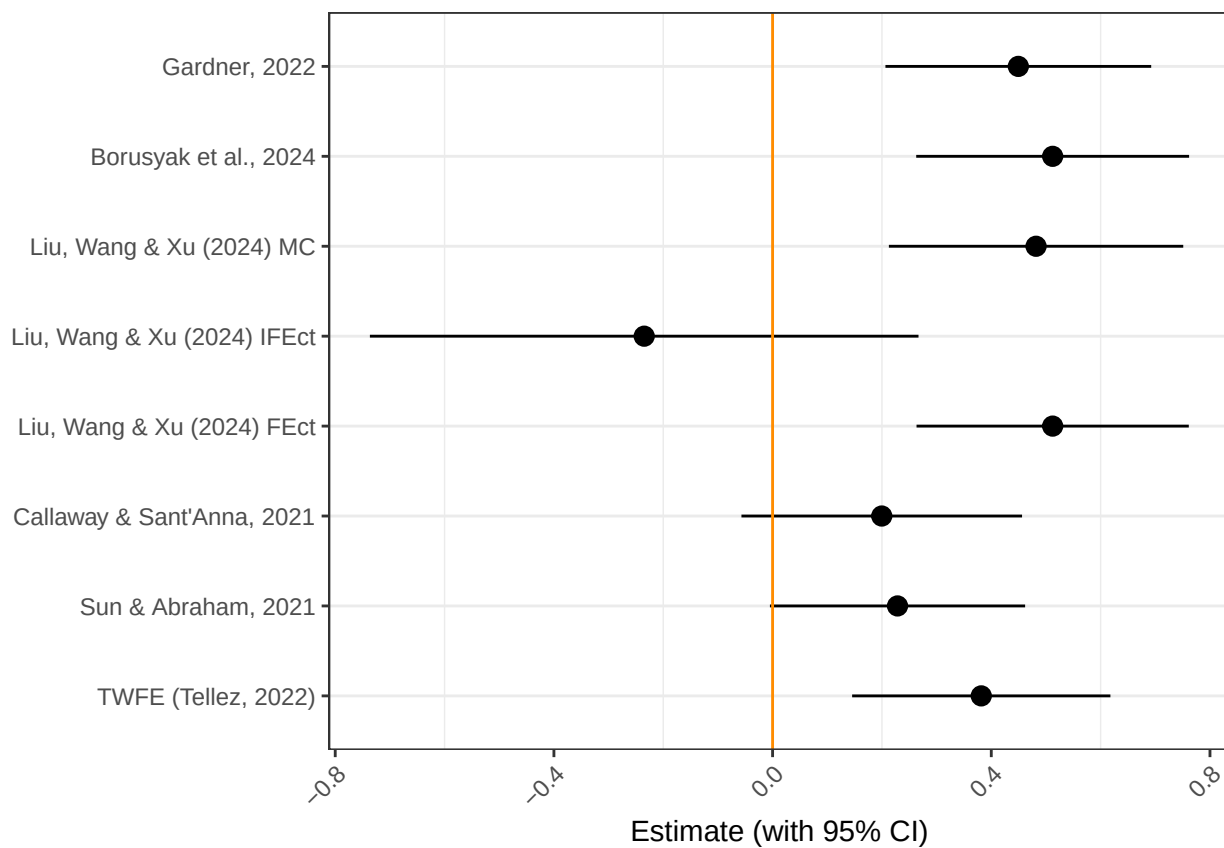


Figure 8: Comparative Plot of Aggregated ATT Estimates Across Estimators

Here, I provide a few ideas that could be carried out in further research which would greatly enhance the causal interpretation of results. First, as mentioned in the limitations, the lack of a treatment variable that takes into consideration the palm-oil production intensity by municipality and year could be severely misleading, and creates significant bias. Data from the Global Oil Palm Dataset 1990-2021 (Descals *et al.*, 2024) includes data on whether the palm-oil plantations were industrial or from smallholders, which may already prove interesting even if not providing a specific production intensity level. Interacting the binary variable with this data would likely provide more interesting results than the current approach interacted with national production statistics.

Another alternative would be getting an estimate of palm-oil planted by municipality (e.g., with geospatial data), and using that measure instead of the national production statistics, as the assumption would shift from parallel production levels across municipalities to linear progression of plantation (i.e., that those with more planted palm trees now had more palm trees planted before too). Such approach has been carried out to measure coca plantations in Colombia by the UNODC’s Illicit Crop Monitoring Programme (ICMP), and should be fairly straightforward to carry out (given the huge plots of land that palm-oil plantations often occupy, which can range from 10,000-25,000 hectares), although, unfortunately, it involves a set of skills which I do not possess.

The reverse causality concerns rightfully highlighted by the author are not solved here and would require of more granular (e.g., monthly) data, or a qualitative analysis of displacement and palm-oil plantation patterns that minimises self-reporting and retrospective bias. The theory that displacement precedes plantation makes more intuitive sense than the one tested for here, which considers plantations as increasing displacement. In Table 5, I show a methodological curiosity, which proves that indeed there are reasons to be concerned about the direction of the causal arrow, given higher displacement rates increase the probability of a municipality having palm-oil plantations, in a binary (logit) model across multiple specifications.

	Model 1	Model 2	Model 3	Model 4
Displacement Rate (IHS)	0.130*** (0.000)	0.359*** (0.000)	0.350*** (0.000)	−0.289*** (0.000)
Coca plantation		−2.182*** (0.000)	−2.295*** (0.000)	−2.166*** (0.000)
FARC Attack			0.786*** (0.000)	0.799*** (0.000)
Displacement x AUC				3.133*** (0.000)
Num. obs.	372	371	371	371
Num. groups: cod_dane	29	29	29	29
Num. groups: year	13	13	13	13
Deviance	0.000	0.000	0.000	0.000
Pseudo R ²	0.829	0.824	0.820	0.816

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 5: Reverse Causality Test – Predicting Palm-Oil Plantation Presence with Displacement and Covariates

Another limitation from both the original paper’s methodology and my own attempts is the potential violation of SUTVA given the likely existence of spillovers given the geographic nature (effect) of displacement and violence. One potential solution may be including a spatial lag term in the model, to explicitly capture how neighbouring (within some distance) towns (displacement rate, violence) affect the outcome of nearby towns over time. Another slightly easier *partial* solution would be to group towns into “blocks” and then estimate block-level treatment effects with cluster-robust SEs, so that within-block spillovers are possible, but between block they are less substantive. Still, this latter partial solution still has some of the same, even if smaller, limitations as the original approach clustering by municipality.

5 References

- Borusyak, K., Jaravel, X. & Spiess, J. (2024). Revisiting Event-Study Designs: Robust and Efficient Estimation. *The Review of Economic Studies*, 91(6), pp.3253-3285.
- Callaway, B. & Sant’Anna, P. (2021). Difference-in-Differences with multiple time periods. *Journal of Econometrics*, 225(2), pp.200-230.
- Chiu, A., Lan, X., Liu, Z. & Xu, Y. (2025, Forthcoming). Causal Panel Analysis under Parallel Trends: Lessons from A Large Reanalysis Study. *American Political Science Review*, Conditionally Accepted. March 24th 2025 version, available here.
- Descals, A., Gaveau, D., Wich, S., Szantoi, Z. & Meijaard, E. (2024). Global mapping of oil palm planting year from 1990 to 2021. *Earth Systems Science Data*, 16, pp.5111-5129.
- Dion, M. & Russler, C. (2008). Eradication Efforts, the State, Displacement and Poverty : Explaining Coca Cultivation in Colombia during Plan Colombia. *Journal of Latin American Studies*, 40, pp.399-421.
- Gardner, J. (2022). Two-stage differences in differences. *arXiv:2207.05943*. [econ.EM]. Available here.
- Liu, L., Wang, Y. & Xu, Y. (2024). A Practical Guide to Counterfactual Estimators for Causal Inference with Time-Series Cross-Sectional Data. *American Journal of Political Science*, 68(1), pp.160-176.
- Rambachan, A. & Roth, J. (2023). A More Credible Approach to Parallel Trends. *The Review of Economic Studies*, 90(5), pp.2555-2591.
- Sun, L. & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2), pp.175-199.
- Tellez, J. (2022). Land, opportunism, and displacement in civil wars: Evidence from Colombia. *American Political Science Review*, 116(2), pp.403-418.

6 Appendix

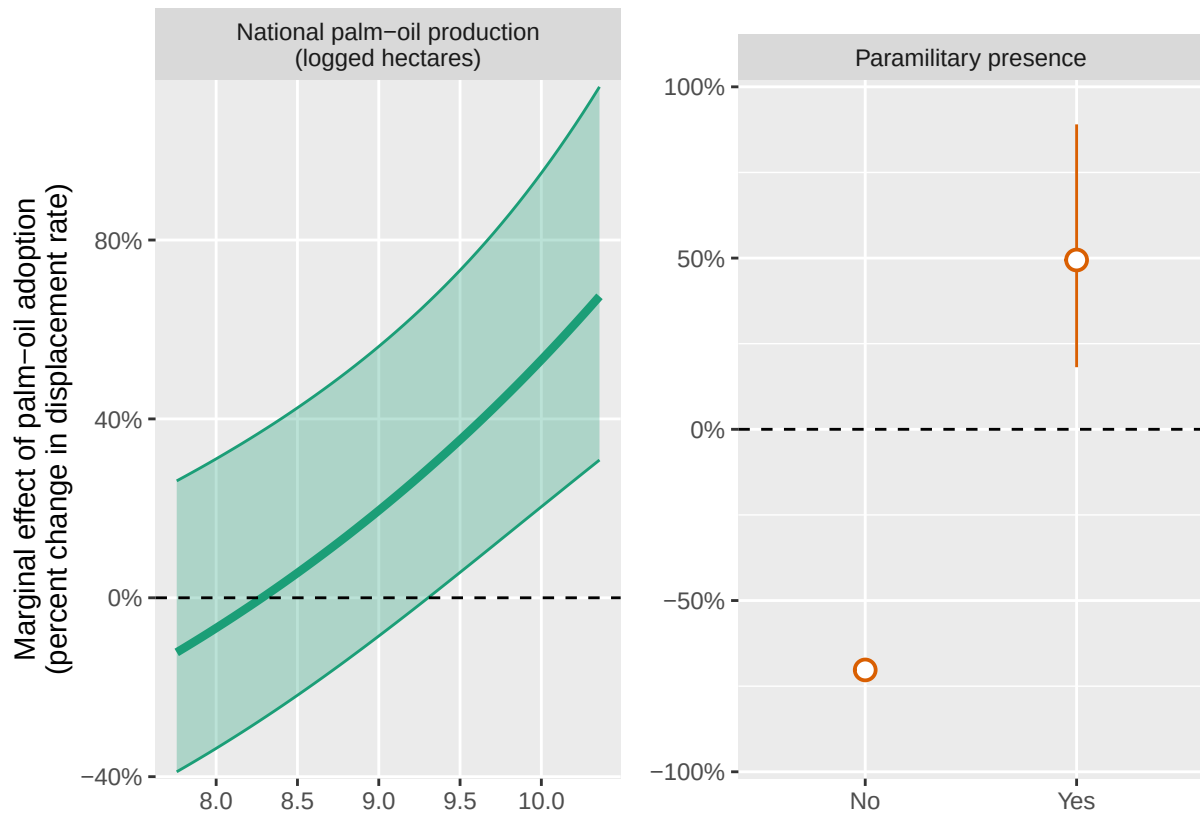


Figure 9: Effect of Palm-Oil Presence on Displacement Rate across National Palm-Oil Production Levels (Left) and Presence/Absence of Paramilitary Groups (Right)

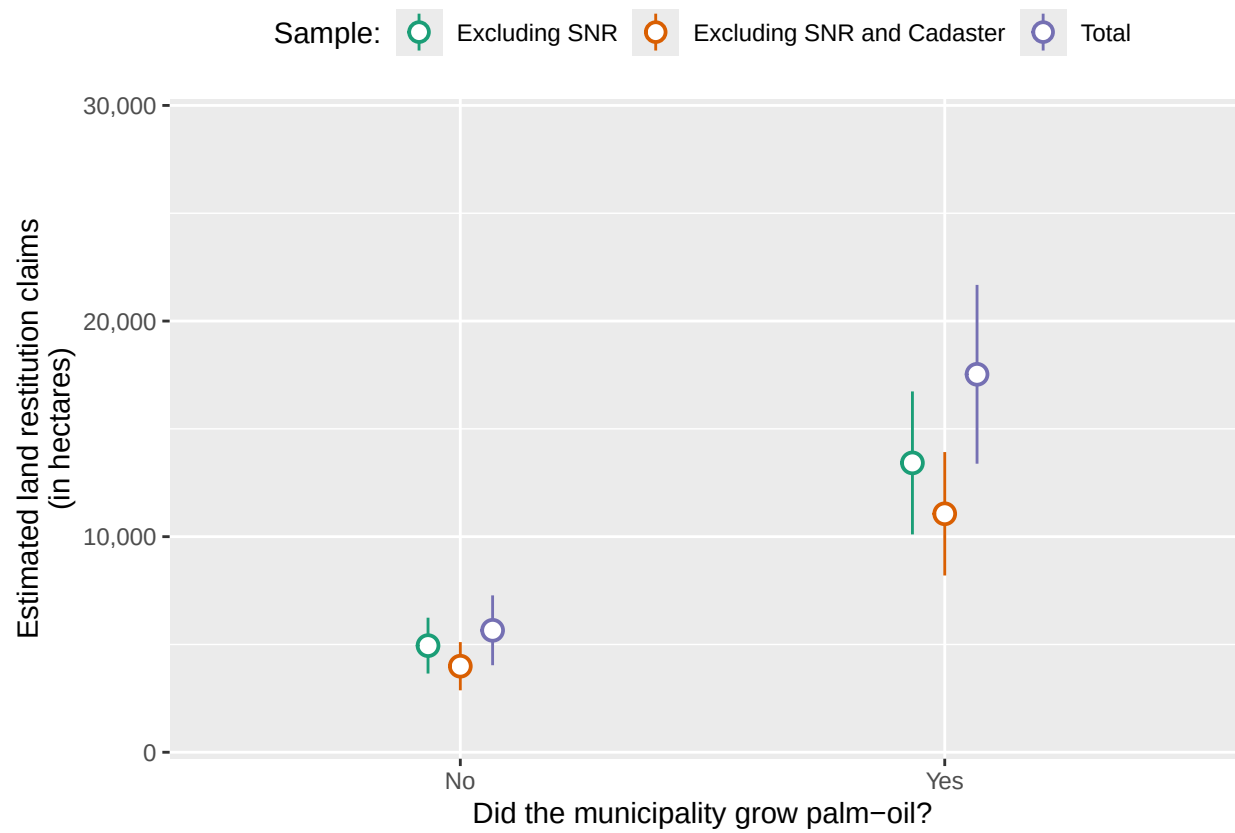


Figure 10: Predicted Amount of Contemporary Land-Restitution Need, Holding other Covariates at the Mean

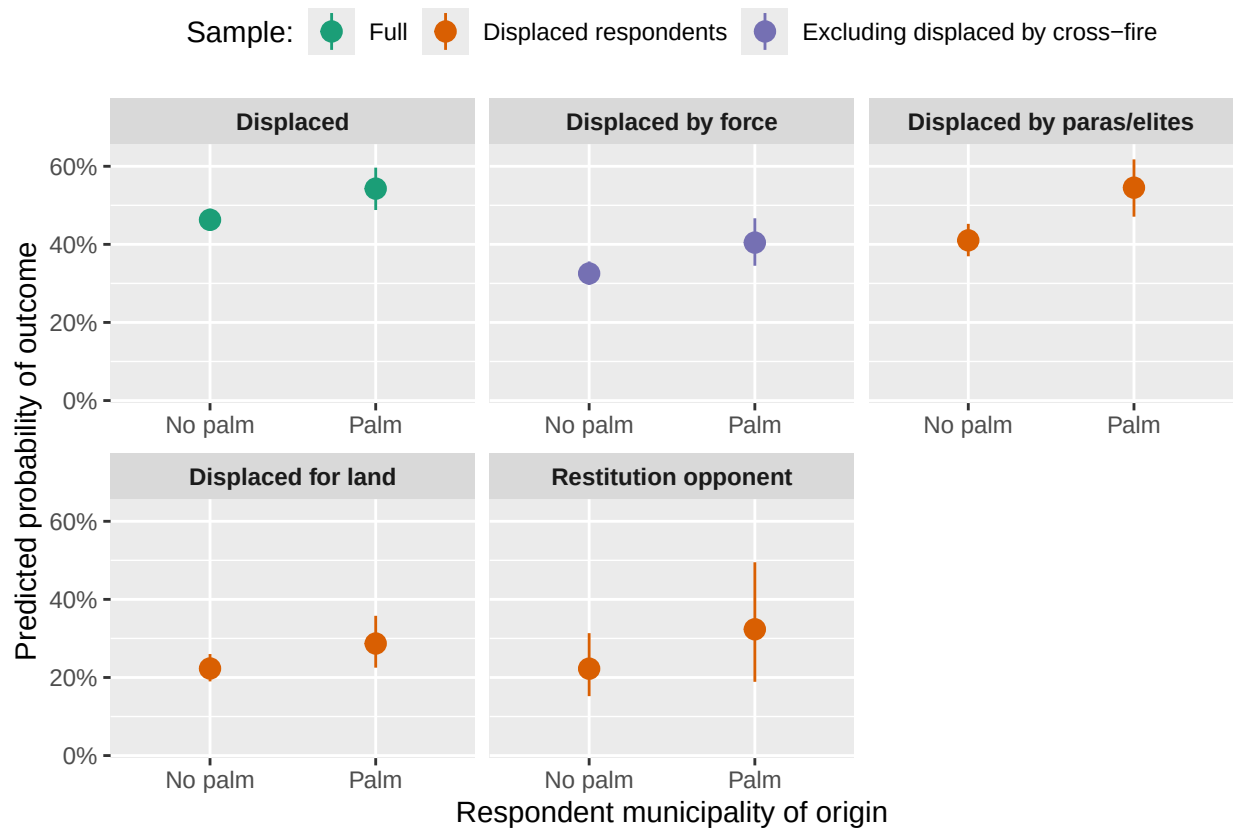


Figure 11: Predicted Probabilities from Logit Models Using Household Survey Data

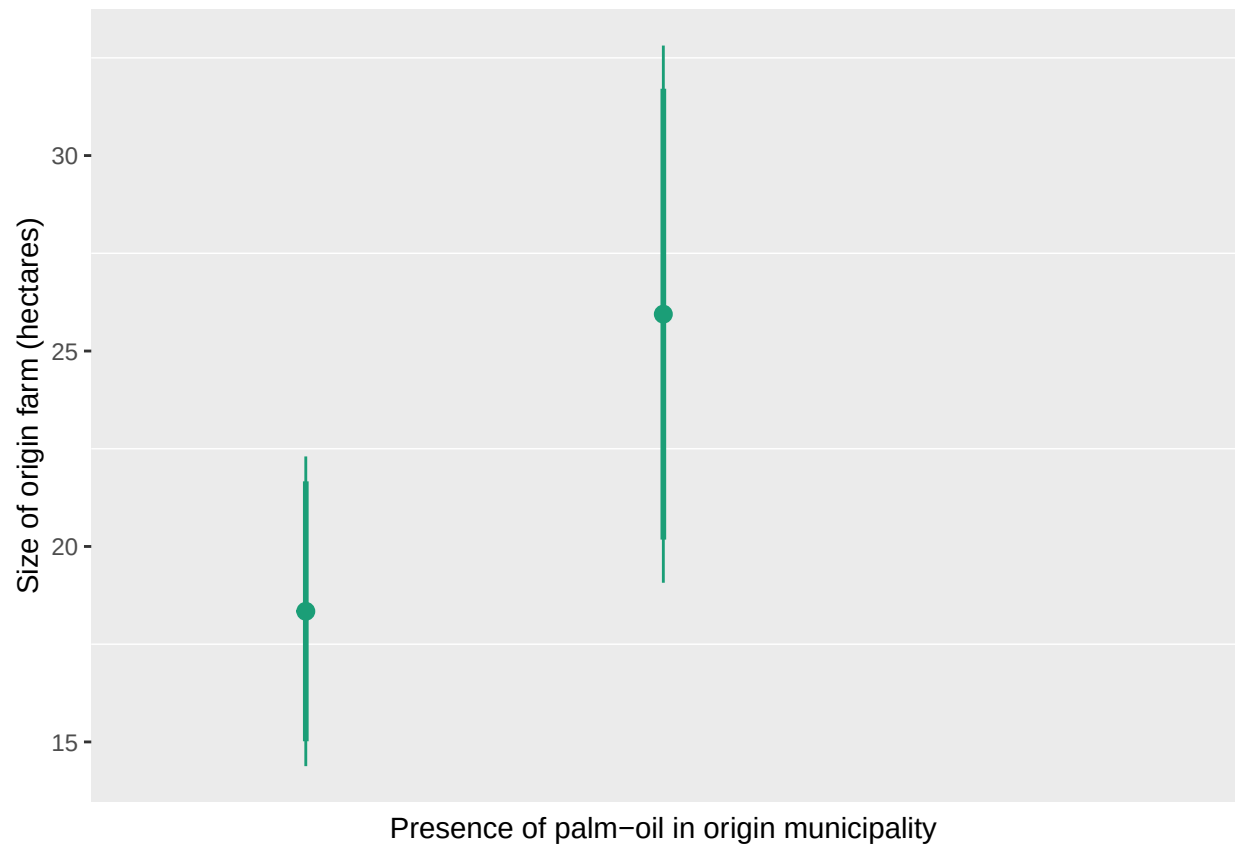


Figure 12: Estimated Amount of Land Owned or Worked Prior to Displacement across Palm-Oil- and Non-Palm-Oil-Growing Municipalities

7 Code appendix

```
# this chunk contains code that sets global options for the entire .Rmd.
# we use include=FALSE to suppress it from the top of the document, but it will still appear in the app
knitr::opts_chunk$set(echo=FALSE, warning=FALSE, message=FALSE, linewidth=60, dev = "pdf")
set.seed(2025) # ensures reproducibility for the full knit

# Core data manipulation and workflow
library(here)
library(tidyverse)

# Individual tidyverse packages
library(dplyr)
library(ggplot2)
library(readr)
library(tidyr)

# Additional visualization packages
library(extrafont)
library(ggrepel)
library(hrbrthemes)
library(paletteer)
library(patchwork)
library(scales)
library(dagitty)
library(ggdag)
library(gridExtra)

# Statistical modeling and analysis
library(broom)
library(fixest)
library(interplot)
library(lfe)
library(did)
library(fect)
library(didimputation)
library(did2s)
library(HonestDiD)
library(ggeffects)

# Results presentation and publication
library(kableExtra)
library(knitr)
library(panelView)
library(texreg)
library(tinytex)

# and any other options in R:
options(scipen=999)

# Change to appropriate working directory
setwd("~/LSE Coursework and Documents/MY457 Causal Inference for Observational and Experimental Studies")
```

```

# Changed name of dataset from df to muni
muni <- read_rds("~/LSE Coursework and Documents/MY457 Causal Inference for Observational and Experimental Data")

#-----TABLE 1-----
# Crosstabulation of AUC and FARC presence
# This is my own original code

tab_farc_auc <- muni %>%
  dplyr::select(cod_dane, auc_dummy, farc_dummy) %>%
  distinct() %>%
  count(auc_dummy, farc_dummy) %>%
  mutate(
    auc_presence = if_else(auc_dummy == 1, "AUC: Yes", "AUC: No"),
    farc_presence = if_else(farc_dummy == 1, "FARC: Yes", "FARC: No")
  ) %>%
  dplyr::select(-auc_dummy, -farc_dummy) %>%
  pivot_wider(
    names_from = farc_presence,
    values_from = n,
    values_fill = 0
  ) %>%
  column_to_rownames("auc_presence")

tab_farc_auc %>%
  kable("latex", caption = "Cross-tabulation of AUC and FARC Presence by Municipality (Unique)") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed", "responsive"),
    full_width = F, position = "center")

# Creating new variable for group presence rather than using distinct dummies
# This allows for testing when both or no groups are present in a municipality
# This is my own original code

muni <- muni %>%
  mutate(group_presence = case_when(
    auc_dummy == 0 & farc_dummy == 0 ~ 0, # neither present
    auc_dummy == 0 & farc_dummy == 1 ~ 1, # FARC only
    auc_dummy == 1 & farc_dummy == 0 ~ 2, # AUC only
    auc_dummy == 1 & farc_dummy == 1 ~ 3  # both present
  ),
  group_presence = factor(group_presence,
    levels = 0:3,
    labels = c("None", "FARC only", "AUC only", "Both"))
  )

#-----FIGURE 1-----
# Rate of Displacement across Municipalities With and Without Palm Oil Cultivations
# Most of the code was copied from the 'modeling-municipal.r' replication script, though some minor changes

update_geom_defaults("label", list(family = "roboto"))
update_geom_defaults("label_repel", list(family = "roboto"))

pDat <- muni %>%
  mutate(year = unclass(year)) %>% # This converts the year variable to numeric given original variable

```

```

mutate(cod_dane = unclass(cod_dane)) %>%
mutate(cod_dane = unclass(sdisp_rate)) %>%
dplyr::select(year, palm_presence, disp_rate) %>%
group_by(year, palm_presence) %>%
summarise(disp_rate = mean(disp_rate, na.rm = TRUE),
          .groups = "drop") %>%

drop_na() %>%
mutate(
  palm_presence = ifelse(palm_presence == 1,
                        "Locations with palm oil",
                        "Locations without palm oil")
)

labels <- pDat %>%
  filter(year == 2003)

# In the original paper he mistakenly indicated that this plot was showing rates per 10,000 residents,
ggplot(pDat,
      aes(x = year, y = disp_rate, color = palm_presence)) +
  geom_line(size = 1.5) +
  theme_ipsum_rc(axis_title_size = 12) +
  labs(x = '',
       y = 'Displacement rate per 1,000 inhabitants',
       color = 'Palm oil cultivations:') +
  theme(legend.position = 'none',
        panel.grid.major = element_blank(),
        text = element_text(family = "roboto")) +
  geom_label_repel(data = labels,
                  aes(x = year, y = disp_rate,
                     label = palm_presence,
                     fill = palm_presence),
                  fontface = "bold",
                  color = "white",
                  nudge_y = 2,
                  size = 5,
                  min.segment.length = Inf,
                  nudge_x = 5) +
  scale_color_brewer(palette = "Dark2") +
  scale_fill_brewer(palette = "Dark2") +
  scale_x_continuous(breaks = seq(1992, 2004, 2))

#-----FIGURE 2-----
# Panelview of palm-oil plantation across municipalities
# This is my own original code

muni <- muni %>%
  mutate(year = unclass(year)) %>% # This converts the year variable to numeric given original variable
  mutate(sdisp_rate = unclass(sdisp_rate))

# A message appears when knitting the document, but I found no guidance online on how to hide it, sorry
panelview(sdisp_rate ~ palm_presence,
          data = muni,
          index = c("cod_dane", "year"),

```

```

    xlab = "Year",
    ylab = "Municipality",
    color = c("darkorange", "seagreen4", "white"),
    background = "white",
    theme.bw = TRUE,
    display.all = TRUE,
    gridOff = TRUE,
    by.timing = TRUE,
    main = "Plantation of Palm-Oil Across Municipalities")

#-----FIGURE 3-----
# Trend in Municipal Palm Oil Presence
# This is my own original code

years_of_interest <- c(1993, 1997, 2001, 2005)

prop_by_year <- muni %>%
  mutate(year = as.integer(year)) %>%
  group_by(year) %>%
  summarize(
    prop = mean(palm_presence == 1, na.rm = TRUE)
  ) %>%
  ungroup()

ggplot(prop_by_year, aes(x = year, y = prop)) +
  geom_line(size = 0.8) +
  geom_point(size = 1.5, alpha = 0.6) +
  geom_point(
    data = filter(prop_by_year, year %in% years_of_interest),
    aes(x = year, y = prop),
    shape = 21,
    fill = "darkorange",
    colour = "black",
    size = 3,
    stroke = 1.2
  ) +
  geom_text(
    data = filter(prop_by_year, year %in% years_of_interest),
    aes(label = percent(prop, accuracy = 0.1)),
    vjust = -1,
    fontface = "bold",
    size = 3
  ) +
  scale_y_continuous(
    labels = percent_format(accuracy = 1),
    expand = expansion(mult = c(0, 0.10))
  ) +
  scale_x_continuous(
    breaks = seq(min(prop_by_year$year), max(prop_by_year$year), by = 2),
    expand = expansion(add = c(0, 0))
  ) +
  labs(
    x = "Year",

```

```

y = "% of Municipalities w/ Palm Oil Plantations",
title = "Trend in Municipal Palm Oil Presence",
subtitle = "With highlighted observations at 1993, 1997, 2001, and 2005"
) +
theme_minimal(base_size = 14) +
coord_cartesian(clip = "off") + # To ensure the figure does not cut the values from the visualisation
theme(
  panel.grid.major.x = element_blank(),
  panel.grid.minor   = element_blank(),
  plot.title         = element_text(face = "bold", size = 14),
  plot.subtitle      = element_text(size = 10),
  plot.margin        = margin(1, 1, 1, 1, "cm"),
  axis.title         = element_text(face = "bold"),
  axis.text          = element_text(colour = "gray30")
)

#-----TABLE 2-----
# Crosstabulation of coca and palm-oil presence across municipalities
# This is my own original code

tab_coca_palm <- muni %>%
  dplyr::select(cod_dane, coca, palm_presence) %>%
  filter(!is.na(coca) & !is.na(palm_presence)) %>%
  distinct() %>%
  count(coca, palm_presence) %>%
  mutate(
    coca_label = if_else(coca == 1, "Coca Plantation: Yes", "Coca Plantation: No", missing = "Coca Plantation: No"),
    palm_label = if_else(palm_presence == 1, "Palm Plantation: Yes", "Palm Plantation: No", missing = "Palm Plantation: No")
  ) %>%
  dplyr::select(-coca, -palm_presence) %>%
  complete(coca_label, palm_label, fill = list(n = 0)) %>%
  pivot_wider(
    names_from = palm_label,
    values_from = n,
    values_fill = 0
  ) %>%
  filter(!is.na(coca_label)) %>%
  column_to_rownames("coca_label")

tab_coca_palm %>%
  kable("latex", caption = "Cross-tabulation of Coca and Palm Plantations by Municipality (Unique)") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed", "responsive"),
    full_width = F, position = "center")

#-----PART OF FIGURE 4-----
# Event study plot with paper's inverse sine outcome.
# This is a mix of paper's code and my own code

onset <- muni %>%
  dplyr::select(cod_dane, year, palm_presence) %>%
  filter(palm_presence == 1) %>%
  group_by(cod_dane) %>%
  summarise(first_treat = min(year))

```

```

muni <- muni %>% # Copied code from author's replication package, just changed name of dataset
  dplyr::select(cod_dane, year, disp_rate, sdisp_rate, ldisp_rate, palm_presence,
    farc_attack, coca, lnatprod, auc_dummy, farc_dummy, group_presence) %>%
  # year treated
  left_join(onset) %>%
  # set never-treated to number above maximum year
  # (any year not in year == never-treated)
  mutate(first_treat = replace_na(first_treat, 10000)) %>%
  # change year format to match suns
  mutate(year_rel = year - 1992,
    first_treat_rel = first_treat - 1992)

sunab_sine <- feols(sdisp_rate ~ sunab(first_treat, year)+farc_attack+coca|as.factor(cod_dane)+as.factor(

sunab_sine_results <- broom::tidy(sunab_sine) %>%
  filter(grepl("year:", term)) %>%
  mutate(time = as.numeric(gsub("year:", "", term))) # Extract numeric time periods

sunab_sine_plot <- ggplot(sunab_sine_results, aes(x = time, y = estimate)) +
  geom_point() +
  geom_errorbar(aes(ymin = estimate - 1.96 * std.error, ymax = estimate + 1.96 * std.error), width = 0.1) +
  geom_vline(xintercept = -1, linetype = "dashed", color = "darkorange") + # Reference period at -1
  geom_hline(yintercept = 0, color = "black")+
  scale_x_continuous(limits = c(-12,10))+
  labs(title = "Inverse-Sine",
    x = "Event Time",
    y = "Coefficient Estimate (Displacement Rate)") +
  theme_bw()

#-----PART OF FIGURE 4-----
# Event study plot with log transformed outcome
# This is my own code

sunab_log <- feols(ldisp_rate ~ sunab(first_treat, year)+farc_attack+coca|as.factor(cod_dane)+as.factor(

sunab_log_results <- broom::tidy(sunab_log) %>%
  filter(grepl("year:", term)) %>%
  mutate(time = as.numeric(gsub("year:", "", term))) # Extract numeric time periods

sunab_log_plot <- ggplot(sunab_log_results, aes(x = time, y = estimate)) +
  geom_point() +
  geom_errorbar(aes(ymin = estimate - 1.96 * std.error, ymax = estimate + 1.96 * std.error), width = 0.1) +
  geom_vline(xintercept = -1, linetype = "dashed", color = "darkorange") + # Reference period at -1
  geom_hline(yintercept = 0, color = "black")+
  scale_x_continuous(limits = c(-12,10))+
  labs(title = "Log-transformed",
    x = "Event Time",
    y = "") +
  theme_bw()

#-----PART OF FIGURE 4-----
# Event study plot with non-transformed (original) outcome
# This is my own code

```



```

sunab_unt <- feols(disprate ~ sunab(first_treat, year)+farc_attack+coca|as.factor(cod_dane)+as.factor(

sunab_unt_results <- broom::tidy(sunab_unt) %>%
  filter(grepl("year::", term)) %>%
  mutate(time = as.numeric(gsub("year::", "", term))) # Extract numeric time periods

sunab_unt_plot <- ggplot(sunab_unt_results, aes(x = time, y = estimate)) +
  geom_point() +
  geom_errorbar(aes(ymin = estimate - 1.96 * std.error, ymax = estimate + 1.96 * std.error), width = 0.1) +
  geom_vline(xintercept = -1, linetype = "dashed", color = "darkorange") + # Reference period at -1
  geom_hline(yintercept = 0, color = "black")+
  scale_x_continuous(limits = c(-12,10))+
  labs(title = "Non-Transformed",
       x = "Event Time",
       y = "",
       caption = "Estimated with Sun & Abraham (2021) estimator. 95% Confidence Intervals shown") +
  theme_bw()

#-----FIGURE 4-----
# Combined event study plots with all three outcomes (inverse-sine; log-transformed; untransformed; resp
# This is my own code

combined_plot <-
  sunab_sine_plot +
  sunab_log_plot +
  sunab_unt_plot +
  plot_layout(ncol = 3, widths = c(1,1,1)) &
  theme(
    plot.margin = margin(5, 5, 5, 5),
    axis.title.x = element_text(size = 10),
    axis.title.y = element_text(size = 10)
  )

print(combined_plot)

#-----FIGURE 5-----
# Placebo Tests for Parallel Trends (with fect package)
# This is my own code

par(mfrow = c(2, 2), mar = c(4, 4, 2, 1)) # Set 2x2 layout

placebo1 <- plot(fect::fect(
  sdisprate ~ palm_presence, data = muni,
  index = c("cod_dane", "year"),
  method = "fe", X = farc_attack + coca,
  se = TRUE, nboots = 100, placeboTest = TRUE, placebo.period = c(-2, 0)
), main = "Placebo Test (ref = -2)", cex.main = 0.9)

placebo2 <- plot(fect::fect(
  sdisprate ~ palm_presence, data = muni,
  index = c("cod_dane", "year"),
  method = "fe", X = farc_attack + coca,
  se = TRUE, nboots = 100, placeboTest = TRUE, placebo.period = c(-3, 0)

```

```

), main = "Placebo Test (ref = -3)", cex.main = 0.9)

placebo3 <- plot(fect::fect(
  sdisp_rate ~ palm_presence, data = muni,
  index = c("cod_dane", "year"),
  method = "fe", X = farc_attack + coca,
  se = TRUE, nboots = 100, placeboTest = TRUE, placebo.period = c(-4, 0)
), main = "Placebo Test (ref = -4)", cex.main = 0.9)

placebo4 <- plot(fect::fect(
  sdisp_rate ~ palm_presence, data = muni,
  index = c("cod_dane", "year"),
  method = "ife", X = farc_attack + coca,
  se = TRUE, nboots = 100, placeboTest = TRUE, placebo.period = c(-5, 0)
), main = "Placebo Test (ref = -5)", cex.main = 0.9)

placebo1 + placebo2
placebo3 + placebo4

par(mfrow = c(1, 1)) # Reset to default
#-----FIGURE 6-----
# Sensitivity Analysis (with honestdid package)
# This is my own code, although some was copied from the HonestDiD documentation, available online at https://www.honestdid.com

# 1. Paste in the extractor function, to extract and aggregate my Sun & Abraham coefficients and variance-covariance matrix
sunab_beta_vcv <- function(sunab_fixest){
  sunab_agg <- sunab_fixest$model_matrix_info$sunab$agg_period
  sunab_names <- names(sunab_fixest$coefficients)
  sunab_sel <- grepl(sunab_agg, sunab_names, perl=TRUE)
  sunab_names <- sunab_names[sunab_sel]
  if(!is.null(sunab_fixest$weights)){
    sunab_wgt <- colSums(sunab_fixest$weights * sign(model.matrix(sunab_fixest)[, sunab_names, drop=FALSE]))
  } else {
    sunab_wgt <- colSums(sign(model.matrix(sunab_fixest)[, sunab_names, drop=FALSE]))
  }
  sunab_cohorts <- as.numeric(gsub(paste0(".*", sunab_agg, ".*"), "\\2", sunab_names, perl=TRUE))
  sunab_mat <- model.matrix(~ 0 + factor(sunab_cohorts))
  sunab_trans <- solve(t(sunab_mat) %*% (sunab_wgt * sunab_mat)) %*% t(sunab_wgt * sunab_mat)
  sunab_coefs <- sunab_trans %*% cbind(sunab_fixest$coefficients[sunab_sel])
  sunab_vcov <- sunab_trans %*% sunab_fixest$cov.scaled[sunab_sel, sunab_sel] %*% t(sunab_trans)
  list(beta = drop(sunab_coefs),
        sigma = sunab_vcov,
        cohorts = sort(unique(sunab_cohorts)))
}

# 2. Extract betahat, variance-covariance matrix, and cohort identifiers for my model:
bv <- sunab_beta_vcv(sunab_sine)
betahat_full <- bv$beta
sigma_full <- bv$sigma
cohorts <- bv$cohorts

# 3. Remove the unwanted coefficients (positions 109 & 110) from covariates:
exclude_pos <- c(109, 110)

```

```

keep_pos      <- setdiff(seq_along(betahat_full), exclude_pos)

betahat <- betahat_full[keep_pos]
sigma   <- sigma_full[keep_pos, keep_pos]

# 4. Set up HonestDiD arguments:
numPre  <- sum(cohorts < 0)
numPost <- sum(cohorts > -1)
l_vec   <- rep(1/numPost, numPost)

kwargs <- list(
  betahat      = betahat,
  sigma        = sigma,
  numPrePeriods = numPre,
  numPostPeriods = numPost,
  l_vec        = l_vec
)

# 5. Compute original CS and relative-magnitudes sensitivity:
original_results <- do.call(HonestDiD::constructOriginalCS,      kwargs)

# Generate sensitivity results over a grid of relative magnitudes (Mbarvec)
sensitivity_results <- do.call(
  HonestDiD::createSensitivityResults_relativeMagnitudes,
  c(kwargs, list(
    Mbarvec      = seq(from = 0.1, to = 0.5, by = 0.1),
    gridPoints    = 100
  ))
)

# 6. Plot:
HonestDiD::createSensitivityPlot_relativeMagnitudes(
  sensitivity_results,
  original_results
)

#-----MODELS FOR TABLES 3 & 4-----
# Models created with feols() function from fixest package
# This is my own code, although the use of covariates and clustering is the same as in original code, w

m1 <- feols(sdisp_rate ~ palm_presence + farc_attack + coca|cod_dane + year, vcov= ~cod_dane, data=muni.
m1a <- feols(ldisp_rate ~ palm_presence + farc_attack + coca|cod_dane + year, vcov= ~cod_dane, data=muni.
m2 <- feols(sdisp_rate ~ palm_presence*lnatprod + farc_attack + coca|cod_dane + year, vcov= ~cod_dane, 
m3 <- feols(sdisp_rate ~ palm_presence*auc_dummy + farc_attack + coca|cod_dane + year, vcov= ~cod_dane, 
m_farc1 <- feols(sdisp_rate ~ palm_presence*farc_dummy + coca|cod_dane + year, vcov= ~cod_dane, data=muni.
m_farc2 <- feols(sdisp_rate ~ palm_presence*farc_dummy + farc_attack + coca|cod_dane + year, vcov= ~cod.

```

```

# My Models
m_x3a <- feols(sdisp_rate ~ palm_presence*auc_dummy*lnatprod + farc_attack + coca|cod_dane + year, vcov=vcov)

m_x3b <- feols(sdisp_rate ~ palm_presence*farc_dummy*lnatprod + farc_attack + coca|cod_dane + year, vcov=vcov)

muni$group_presence <- relevel(factor(muni$group_presence), ref = "None") # So I can recover coefficients
m4 <- feols(sdisp_rate ~ palm_presence*group_presence + farc_attack + coca|cod_dane + year, vcov=vcov)

m4_alt <- feols(sdisp_rate ~ palm_presence*auc_dummy + palm_presence*farc_dummy + farc_attack + coca|cod_dane + year, vcov=vcov)

#-----TABLE 3-----
# Created with texreg(), copied in-text below for proper rendering. The paper uses stargazer() for his
# This is my own code

texreg(l = list(m1,m2,m5,m_farc1,m_farc2),
custom.model.names = c("Model 1","Model 2","Model 3","Model 4", "Model 4+"),
custom.coef.names = c("Palm-oil plantation","FARC Attack","Coca plantation","Plantation x Natl Prod", "Plantation x Natl Prod"),
digits = 3,
include.nobs = T,
include.rsquared = F,
include.adjrs = F,
include.aic = F,
include.bic = F,
include.loglik = F)

#-----TABLE 4-----
# Created with texreg(), copied in-text below for proper rendering. The paper uses stargazer() for his
# This is my own code

texreg(l = list(m_x3a,m_x3b, m4, m4_alt),
custom.header = list("Triple Interaction" = 1:2, "Group Presence" = 3, "Multiple Dummies" = 4),
custom.model.names = c("Model 1","Model 2","Model 3 (ref=None)","Model 4"),
custom.coef.names = c("Palm-oil plantation","FARC Attack","Coca plantation","Plantation x AUC", "Plantation x AUC"),
digits = 3,
include.nobs = T,
include.rsquared = F,
include.adjrs = F,
include.aic = F,
include.bic = F,
include.loglik = F)

#-----MODERN ESTIMATOR MODEL-----
# This code uses Callaway & Sant'Anna's (2021) estimator, results presented in Figure 6.
# This is my own code

muni_cs_did <- muni %>%
  mutate(cod_dane = as.numeric(cod_dane))
muni_cs_did$first_treat[muni_cs_did$first_treat > 5000] <- 0 # Recode to meet control group value in cs

cs_sine <- att_gt(yname = "sdisp_rate",
  tname = "year",
  idname = "cod_dane",
  gname = "first_treat",

```

```

        control_group="nevertreated",
        clustervars = "cod_dane",
        base_period = "universal",
        allow_unbalanced_panel = TRUE,
        data = muni_cs_did
    )

agg.agt <- aggte(cs_sine, type = "simple", na.rm=T) # ATT: 0.1997, 95% CI: [-0.0541,0.4534]

#-----MODERN ESTIMATOR EVENT STUDY-----
# This code uses Callaway & Sant'Anna's (2021) estimator, and plots an event-study plot of its results.
# This is my own code

agg.es <- aggte(cs_sine, type = "dynamic",na.rm=T)
ggdid(agg.es, title = "Callaway & Sant'Anna (2021) Event Study Plot", ylab = "Coefficient Estimate (Displacement Rate)",
      #-----MODERN ESTIMATOR GROUP-COHORT EFFECTS-----
      # This code uses Callaway & Sant'Anna's (2021) estimator, and plots an the group-cohort aggregated ATTs
      # This is my own code

agg.gs <- aggte(cs_sine, type = "group",na.rm=T)
ggdid(agg.gs, title = "Callaway & Sant'Anna (2021) Group Aggregated Coefficients")

#-----FIGURE 7-----
# Event-study plots of FEct, IFect and MC estimators from Liu, Wang and Xu (2024) (with fect package)
# This is my own code

# FEct estimator
FEct_1 <- fect(sdisp_rate ~ palm_presence, data = muni, index = c("cod_dane","year"),
               X = farc_attack+coca,
               force = "two-way", method = "fe", CV = TRUE, r = c(0, 5),
               se = TRUE, nboots = 200, parallel = TRUE) # default number of bootstraps
FEct_plot <- plot(FEct_1, main = "", ylab = "Coefficient Estimate (Displacement Rate)", xlab = "",
                 cex.main = 0.8, cex.lab = 0.8, cex.axis = 0.8)

# IFect estimator
IFect_2 <- fect(sdisp_rate ~ palm_presence, data = muni, index = c("cod_dane","year"),
               X = farc_attack+coca,
               force = "two-way", method = "ife", CV = TRUE, r = c(0, 5),
               se = TRUE, nboots = 200, parallel = TRUE)
IFect_plot <- plot(IFect_2, main = "Estimated ATT (FEct; IFect; MC) - Liu, Wang and Xu (2024) estimator",
                 cex.main = 0.8, cex.lab = 0.8, cex.axis = 0.8)

# MC estimator
MC_3 <- fect(sdisp_rate ~ palm_presence, data = muni, index = c("cod_dane","year"),
             X = farc_attack+coca,
             force = "two-way", method = "mc", CV = TRUE, r = c(0, 5),
             se = TRUE, nboots = 200, parallel = TRUE)
MC_plot <- plot(MC_3, main = "", ylab = "", xlab = "",
               cex.main = 0.8, cex.lab = 0.8, cex.axis = 0.8)

par(mfrow = c(1, 3), mar = c(4, 4, 2, 1))

```

```

FEct_plot + IFect_plot + MC_plot

par(mfrow = c(1, 1)) # reset

#-----MODERN ESTIMATOR-----
# This code uses Borusyak et al.'s (2024) estimator, and estimates an aggregated ATT, which can be seen
# This is my own code

# package developed by Borusyak et al., 2021; 2024
muni <- muni %>% # Create cohort/group variable, coded 0 for never treated
  group_by(cod_dane) %>%
  mutate(treat_group = ifelse(palm_presence == 1, year, NA)) %>%
  fill(treat_group, .direction = "downup") %>%
  mutate(treat_group = ifelse(is.na(treat_group), 0, treat_group))

borusyak_imputation <- did_imputation(
  data      = muni,
  yname     = "sdisp_rate",
  idname    = "cod_dane",
  tname     = "year",
  gname     = "treat_group"
)

#-----MODERN ESTIMATOR-----
# This code uses Gardner's (2022) estimator, and estimates an aggregated ATT, which can be seen in Figure
# This is my own code

did2s <- did2s(
  data = muni,
  yname = "sdisp_rate",
  first_stage = ~ farc_attack + coca | cod_dane + year,
  second_stage = ~ i(palm_presence),
  treatment = "palm_presence",
  cluster_var = "cod_dane"
)

#-----DATA MANIPULATION FOR FIGURE 8-----
# Here I extract coefficients from the different estimators used, and put them in the same dataframe for
# This is my own code

# First, I extract the coefficients from all my models
# TWFE (Tellez, 2022)
df_twfe <- data.frame(
  estimate = m1$coefficients,
  std.error = m1$se,
  model = "TWFE (Tellez, 2022)"
)
df_twfe <- df_twfe[1,]

# Sun & Abraham, 2021
sunab_sine_att <- aggregate(sunab_sine, agg = "att")
df_sa <- data.frame(
  estimate = sunab_sine_att

```

```

)
df_sa <- df_sa %>%
  mutate(estimate = df_sa$estimate.Estimate,
         std.error = df_sa$estimate.Std..Error,
         model = "Sun & Abraham, 2021")

# Callaway & Sant'Anna, 2021
df_csa <- data.frame(
  estimate = agg.agt$overall.att,
  std.error = agg.agt$overall.se,
  model = "Callaway & Sant'Anna, 2021"
)

# Liu, Wang & Xu, 2024 (FEct) - Weigh observations equally (est.avg) as some units have less observations
df_fe <- data.frame(
  estimate = FEct_1$est.avg,
  model = "Liu, Wang & Xu (2024) FEct"
)
df_fe <- df_fe %>%
  mutate(estimate = df_fe$estimate.ATT.avg,
         std.error = df_fe$estimate.S.E.)

# Liu, Wang & Xu, 2024 (IFEct) - Weigh observations equally (est.avg) as some units have less observations
df_ife <- data.frame(
  estimate = IFEct_2$est.avg,
  model = "Liu, Wang & Xu (2024) IFEct"
)
df_ife <- df_ife %>%
  mutate(estimate = df_ife$estimate.ATT.avg,
         std.error = df_ife$estimate.S.E.)

# Liu, Wang & Xu, 2024 (MC) - Weigh observations equally (est.avg) as some units have less observations
df_mc <- data.frame(
  estimate = MC_3$est.avg,
  model = "Liu, Wang & Xu (2024) MC"
)
df_mc <- df_mc %>%
  mutate(estimate = df_mc$estimate.ATT.avg,
         std.error = df_mc$estimate.S.E.)

# Borusyak et al., 2024
df_borusyak <- data.frame(
  estimate = borusyak_imputation$estimate,
  std.error = borusyak_imputation$std.error,
  model = "Borusyak et al., 2024"
)

# Gardner, 2022
df_gardner <- tidy(did2s, conf.int = TRUE) %>%
  transmute(term = term, estimate = estimate, std.error = std.error,
            model = "Gardner, 2022")
df_gardner <- df_gardner[2,]

```

```

# bind all the dfs into one
df_all <- bind_rows(
  df_twfe,
  df_sa,
  df_csa,
  df_fe,
  df_ife,
  df_mc,
  df_borusyak,
  df_gardner
) %>%
  # compute 95% CI bounds
  mutate(
    lower = estimate - 1.96 * std.error,
    upper = estimate + 1.96 * std.error,
    # set the order of models on the x-axis:
    model = factor(model, levels = c(
      "TWFE (Tellez, 2022)",
      "Sun & Abraham, 2021",
      "Callaway & Sant'Anna, 2021",
      "Liu, Wang & Xu (2024) FEct",
      "Liu, Wang & Xu (2024) IFect",
      "Liu, Wang & Xu (2024) MC",
      "Borusyak et al., 2024",
      "Gardner, 2022"
    ))
  )

#-----FIGURE 8-----
# Visualisation of all the ATTs estimated with a variety of estimators, including the paper's original
# This is my own code

# plot
ggplot(df_all, aes(x = model, y = estimate)) +
  geom_pointrange(aes(ymin = lower, ymax = upper), size = 0.6) +
  coord_flip() +
  labs(
    x = NULL,
    y = "Estimate (with 95% CI)"
  ) +
  geom_hline(yintercept = 0.0, linetype = "solid", color = "darkorange") +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    panel.grid.major.x = element_blank()
  )

#-----TABLE 5-----
# This code creates four different binary (logit) models testing whether there is evidence to be concerned
# This is my own code

rc1 <- feglm(palm_presence ~ sdisp_rate|cod_dane+year, data = muni, family = "binomial", vcov = ~cod_dane)

```



```

rc2 <- feglm(palm_presence ~ sdisp_rate+coca|cod_dane+year, data = muni, family = "binomial", vcov = ~c
rc3 <- feglm(palm_presence ~ sdisp_rate+coca+farc_attack|cod_dane+year, data = muni, family = "binomial

muni$group_presence <- relevel(factor(muni$group_presence), ref = "None")
rc4 <- feglm(palm_presence ~ sdisp_rate*group_presence+coca+farc_attack|cod_dane+year, data = muni, fam

texreg(l = list(rc1,rc2,rc3,rc4),
custom.model.names = c("Model 1","Model 2","Model 3","Model 4"),
custom.coef.names = c("Displacement Rate (IHS)","Coca plantation","FARC Attack","Displacement x AUC"),
digits = 3,
include.nobs = T,
include.rsquared = F,
include.adjrs = F,
include.aic = F,
include.bic = F,
include.loglik = F)

# Read the dataset again to match name used in replication code
df <- read_rds("~/LSE Coursework and Documents/MY457 Causal Inference for Observational and Experimental

#-----PAPER'S FIGURE 2-----
# Copied code from authors 'modeling-municipal.r' file, made minor changes to solve errors when running

df <- df %>%
  mutate(sdisp_rate = unclass(sdisp_rate),
         palm_presence = unclass(palm_presence),
         lnatprod = unclass(lnatprod),
         farc_attack = unclass(farc_attack),
         coca = unclass(coca),
         year = unclass(year),
         cod_dane = unclass(cod_dane))

m2 = felm(sdisp_rate ~ palm_presence*lnatprod + farc_attack + coca - lnatprod
          | cod_dane + year | 0 | cod_dane, exactDOF = 'rM',
          data = df)

m5 = felm(sdisp_rate ~ palm_presence*auc_dummy + farc_attack + coca - auc_dummy
          | cod_dane + year | 0 | cod_dane, exactDOF = 'rM',
          data = df)

# marginal effects
## code from Patrick Baylis to deal with felm: https://www.patrickbaylis.com/blog/2021-01-22-predict-pa
predict_partial <- function(object, newdata, se.fit = FALSE,
                           interval = "none",
                           level = 0.95){
  if(missing(newdata)) {
    stop("predict_partial requires newdata and predicts for all group effects = 0.")
  }

  newdata <- newdata; se.fit = T; interval = "confidence"; level = 0.95

```

```

# Extract terms object, removing response variable
tt <- terms(object)
Terms <- delete.response(tt)

# Remove intercept
attr(Terms, "intercept") <- 0

X <- model.matrix(Terms, data = newdata)

if (class(object) == "fixest") {
  B <- as.numeric(coef(object))
  df <- attributes(vcov(fit_feols, attr = T))$dof.K
} else if (class(object) %in% c("lm", "felm")) {
  B <- as.numeric(object$coef)
  df <- object$df.residual
} else {
  stop("class(object) should be lm, fixest, or felm.")
}

fit <- data.frame(fit = as.vector(X %*% B))

if(se.fit | interval != "none") {
  sig <- vcov(object)
  se <- apply(X, MARGIN = 1, FUN = get_se, sig = sig)
}

if(interval == "confidence"){
  t_val <- qt((1 - level) / 2 + level, df = df)
  fit$lwr <- fit$fit - t_val * se
  fit$upr <- fit$fit + t_val * se
} else if (interval == "prediction"){
  stop("interval = \"prediction\" not yet implemented")
}

if(se.fit){
  return(list(fit=fit, se.fit = se))
} else {
  return(fit)
}
}

get_se <- function(r, sig) {
  # Compute linear combination, helper function for predict_partial
  # Given numeric vector r (the constants) and vcov sig (the ), compute SE
  r <- matrix(r, nrow = 1)
  sqrt(r %*% sig %*% t(r))
}

# new data for lnatprod
newdata = tibble(palm_presence = 1,
                 farc_attack = 0,
                 coca = 0,
                 lnatprod = seq(min(df$lnatprod, na.rm = TRUE),

```

```

max(df$lnatprod, na.rm = TRUE),
by = .05))

# get marginal effects
preds = predict_partial(m2, newdata = newdata,
                        se.fit = TRUE)

preds = bind_cols(newdata, preds) %>%
  # exponentiate to approximate change
  mutate(effect = (exp(fit) - 1),
         low = (exp(fit - 1.96*se.fit) - 1),
         hi = (exp(fit + 1.96*se.fit) - 1)) %>%
  mutate(group = "National palm-oil production\n(logged hectares)")

p1 = ggplot(preds, aes(x = lnatprod, y = effect, ymin = low,
                      ymax = hi)) +
  geom_ribbon(alpha = .3, color = "#1b9e77", fill = "#1b9e77") +
  geom_line(size = 1.5, color = "#1b9e77") +
  labs(x = NULL,
       y = "Marginal effect of palm-oil adoption\n(percent change in displacement rate)") +
  scale_y_percent() +
  facet_wrap(vars(group)) +
  theme(panel.grid.minor = element_blank()) +
  geom_hline(yintercept = 0, lty = 2)

#-----PAPER'S FIGURE 2-----
# Copied code from authors 'modeling-municipal.r' file, made minor changes to solve errors when running

# new data for auc
newdata = tibble(palm_presence = 1,
                 farc_attack = 0,
                 coca = 0,
                 auc_dummy = c(0, 1))

# get marginal effects
preds = predict_partial(m5, newdata = newdata,
                        se.fit = TRUE)

preds = bind_cols(newdata, preds) %>%
  # exponentiate to approximate change
  mutate(effect = (exp(fit) - 1),
         low = (exp(fit - 1.96*se.fit) - 1),
         hi = (exp(fit + 1.96*se.fit) - 1)) %>%
  mutate(group = "Paramilitary presence")

p2 = ggplot(preds, aes(x = factor(auc_dummy), y = effect,
                      ymin = low,
                      ymax = hi)) +
  geom_pointrange(fatten = 1, size = 3, shape = 21, fill = "white",

```

```

        position = position_dodge(width = .2),
        color = "#d95f02") +
labs(x = NULL,
     y = NULL) +
scale_y_percent(limits = c(-1, 1)) +
scale_x_discrete(breaks = c(0, 1),
                 labels = c("No", "Yes")) +
facet_wrap(vars(group)) +
geom_hline(yintercept = 0, lty = 2)

p1 + p2

#-----PAPER'S FIGURE 3-----
# Copied code from authors 'fondo-analysis.r' file, made minor changes to solve errors when running code

df2 <- readRDS("~/LSE Coursework and Documents/MY457 Causal Inference for Observational and Experimental Data")

inv_sine = function(x)
{
  log(x + sqrt((x^2 + 1)))
}

m1_total = lm(area_restitucion_total ~ palm + nbi + coca + disbogota +
              farc + auc + area_cartografica + rural + log(altitude) +
              log(pib_percapita) + l_area, data = df2)

m1_nosnr = lm(area_restitucion_nosnr ~ palm + nbi + coca + disbogota +
              farc + auc + area_cartografica + rural + log(altitude) +
              log(pib_percapita) + l_area, data = df2)

m1_none = lm(area_restitucion_nosnr_nocatast ~ palm + nbi + coca + disbogota +
              farc + auc + area_cartografica + rural + log(altitude) +
              log(pib_percapita) + l_area, data = df2)

m2 = lm(inv_sine(area_restitucion_total) ~ palm + nbi + coca + disbogota +
        farc + auc + area_cartografica + rural + log(altitude) +
        log(pib_percapita) + l_area, data = df2)

# get predictions
get_preds = function(x, sample){
  pred = ggeffect(x, term = "palm [0,1]") %>%
    as_tibble() %>%
    mutate(samp = sample)
  return(pred)
}

preds = bind_rows(get_preds(m1_total, "Total"),
                  get_preds(m1_nosnr, "Excluding SNR"),
                  get_preds(m1_none, "Excluding SNR and Cadaster")) %>%
# get back on original scale
mutate(across(c(predicted, conf.low, conf.high), .fns = ~.*1e3))

```

```

# plot
ggplot(preds, aes(x = factor(x), y = predicted,
                    ymin = conf.low, ymax = conf.high,
                    color = samp)) +
  geom_pointrange(fatten = 1, size = 3, shape = 21, fill = "white",
                  position = position_dodge(width = .2)) +
  labs(y = "Estimated land restitution claims\n(in hectares)",
       x = "Did the municipality grow palm-oil?",
       color = "Sample:") +
  scale_x_discrete(breaks = c(0, 1), labels = c("No", "Yes")) +
  scale_y_comma(limits = c(0, 30000)) +
  theme(legend.position = "top") +
  guides(colour = guide_legend(override.aes = list(size = .7))) +
  scale_color_brewer(palette = "Dark2")

#-----PAPER'S FIGURE 4-----
# Copied code from authors 'modeling-survey.r' file, made minor changes to solve errors when running co

df3 <- readRDS("~/LSE Coursework and Documents/MY457 Causal Inference for Observational and Experimental

# Survey: palm effects -----

## models
m1 = glm(displace ~ palmDummy + age + ed + child + sex + hhsize,
         family = binomial(link = 'logit'), data = df3)

m2 = glm(displace_force2 ~ palmDummy + age + ed + child + sex + hhsize,
         family = binomial(link = 'logit'), data = df3)

m3 = glm(displace_land ~ palmDummy + age + ed + child + sex + hhsize,
         family = binomial(link = 'logit'),
         data = df3[df3$displace == 1,])

m4 = glm(displace_paras ~ palmDummy + age + ed + child + sex + hhsize,
         family = binomial(link = 'logit'),
         data = df3[df3$displace == 1,])

m5 = lm(land_owned ~ palmDummy + age + ed + child + sex + hhsize,
        data = df3[df3$displace == 1,])

m6 = glm(claimant ~ palmDummy + age + ed + child + sex + hhsize,
         family = binomial(link = 'logit'),
         data = df3[df3$displace == 1,])

# predicted probabilities
preds = tibble(outcomes = c("Displaced", "Displaced by force",
                             "Displaced for land", "Displaced by paras/elites",
                             "Restitution opponent"),
               models = list(m1, m2, m3, m4, m6)) %>%

```

```

mutate(preds = map(models, ~ggeffect(.x, terms = "palmDummy"))) %>%
dplyr::select(outcomes, preds) %>%
unnest(preds) %>%
mutate(sample = case_when(outcomes == "Displaced" ~ "Full",
                           outcomes == "Displaced by force" ~ "Excluding displaced by cross-fire",
                           TRUE ~ "Displaced respondents")) %>%
mutate(sample = fct_relevel(sample, "Full")) %>%
mutate(palm = ifelse(x == 1, "Palm", "No palm"))

ggplot(preds, aes(x = palm, y = predicted,
                  ymin = conf.low, ymax = conf.high,
                  color = sample)) +
  geom_pointrange(size = .65) +
  facet_wrap(vars(outcomes), scales = "free_x") +
  labs(x = "Respondent municipality of origin", color = "Sample:",
       y = "Predicted probability of outcome") +
  scale_color_brewer(palette = "Dark2") +
  theme(legend.position = "top",
        strip.text = element_text(face = "bold")) +
  scale_y_percent(limits = c(0, .65))

#-----PAPER'S FIGURE 5-----
# Copied code from authors 'modeling-survey.r' file, made minor changes to solve errors when running co

# land-owned
pDat = ggeffect(m5, 'palmDummy') %>%
  mutate(lo90 = predicted - qnorm(.95)*`std.error`,
         hi90 = predicted + qnorm(.95)*`std.error`,
         lo95 = predicted - qnorm(.975)*`std.error`,
         hi95 = predicted + qnorm(.975)*`std.error`)

ggplot(pDat) +
  geom_linerange(aes(x = factor(x), ymin = lo95, ymax = hi95),
                lwd = .5, color = "#1b9e77") +
  geom_pointrange(aes(x = factor(x), y = predicted,
                      ymin = lo90, ymax = hi90), lwd = 1,
                  color = "#1b9e77") +
  labs(x = 'Presence of palm-oil in origin municipality',
       y = 'Size of origin farm (hectares)') +
  theme(legend.position = 'top',
        axis.ticks.x = element_blank(),
        panel.grid.major = element_blank(),
        axis.text.x = element_blank())

# this chunk generates the complete code appendix.
# eval=FALSE tells R not to re-run (``evaluate'') the code here.

```